# INTRODUCTION TO ARTIFICIAL INTELLIGENCE IN HEALTHCARE

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# During This Course, You Will Learn

- The basic ideas behind Artificial Intelligence (AI) and Machine Learning (ML)
- Common terms and concepts used in AI/ML
- Different types of medical data used in AI applications
- How to set realistic expectations about what AI can do in healthcare
- How to communicate effectively with computer science and data experts

# Upcoming Lectures

- **Introduction to AI and How Data Shapes Modern ML.**
- Overview of Machine Learning and Deep Learning Algorithms.
- LLMs Are Awesome!
- How to read an AI paper and evaluate an AI product.
- AI Applications in Medicine and Current Research at the University of Alberta.

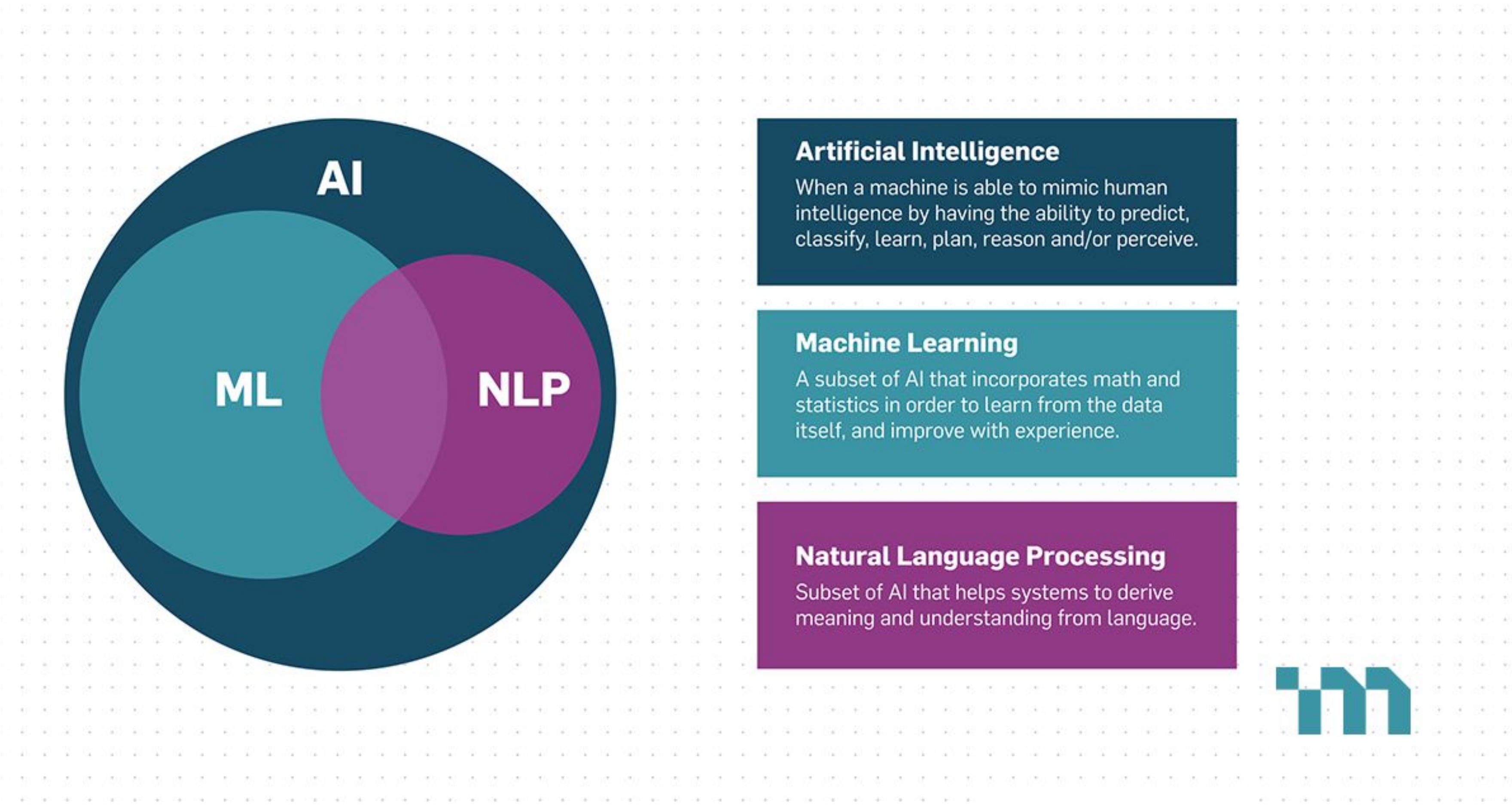
# Artificial Intelligence in Our Life





# Artificial Intelligence

- Artificial intelligence (AI) are machines that can mimic the cognitive functions of humans to perform tasks of problem-solving and learning.



# Brief History of AI

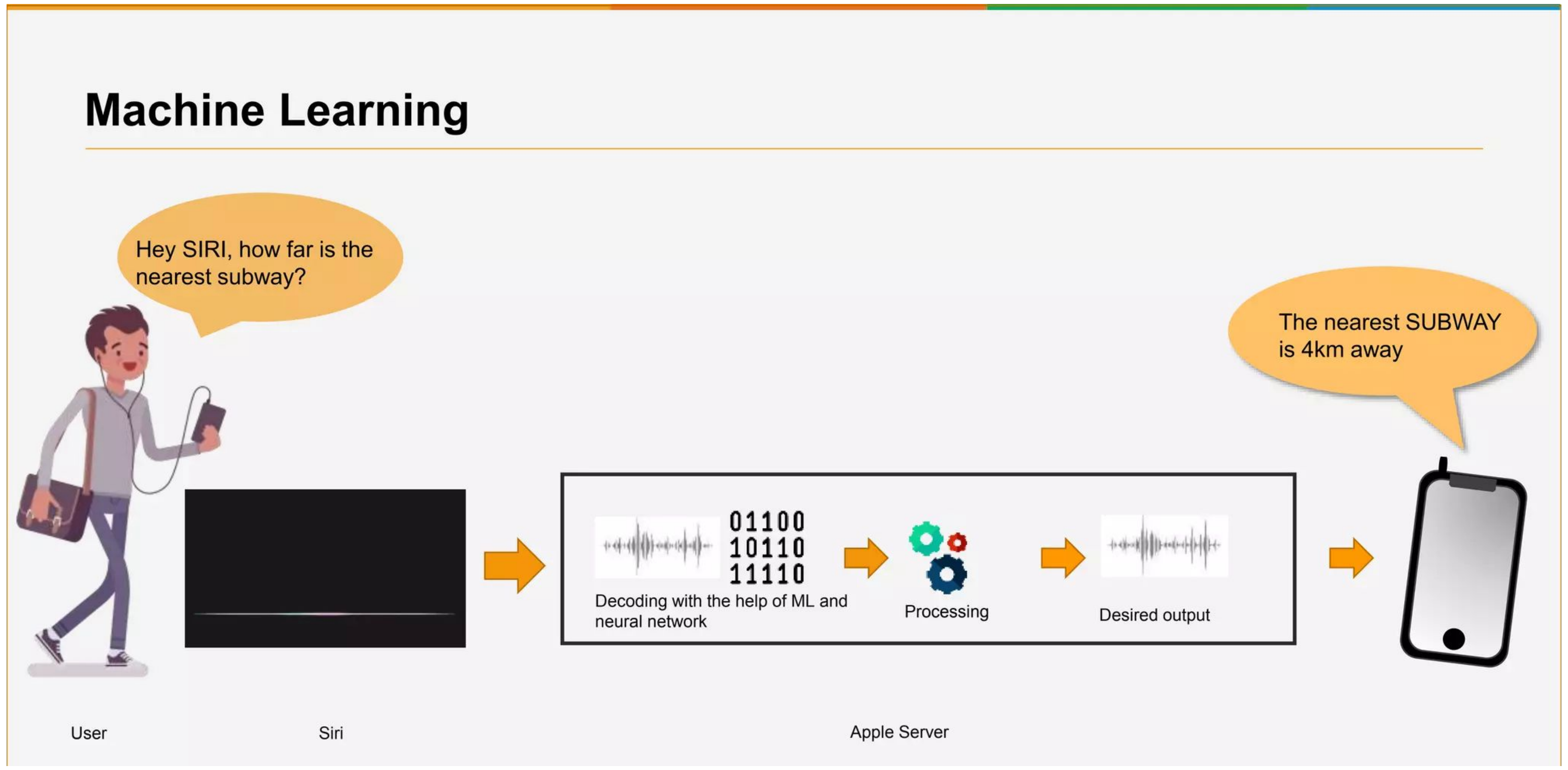
- 1943- McCulloch and Pitts propose the first model of artificial neurons.
- 1950- Alan Turing introduces the Turing test.
- 1956- Term “Artificial Intelligence” coined at the Dartmouth Conference.
- 1980s- Rise of expert systems like DENDRAL.
- 1997- IBM’s Deep Blue defeats chess champion, Garry Kasparov.
- 2000s- Advances in machine learning and neural networks fuel modern AI applications.

# AI in Medical Field

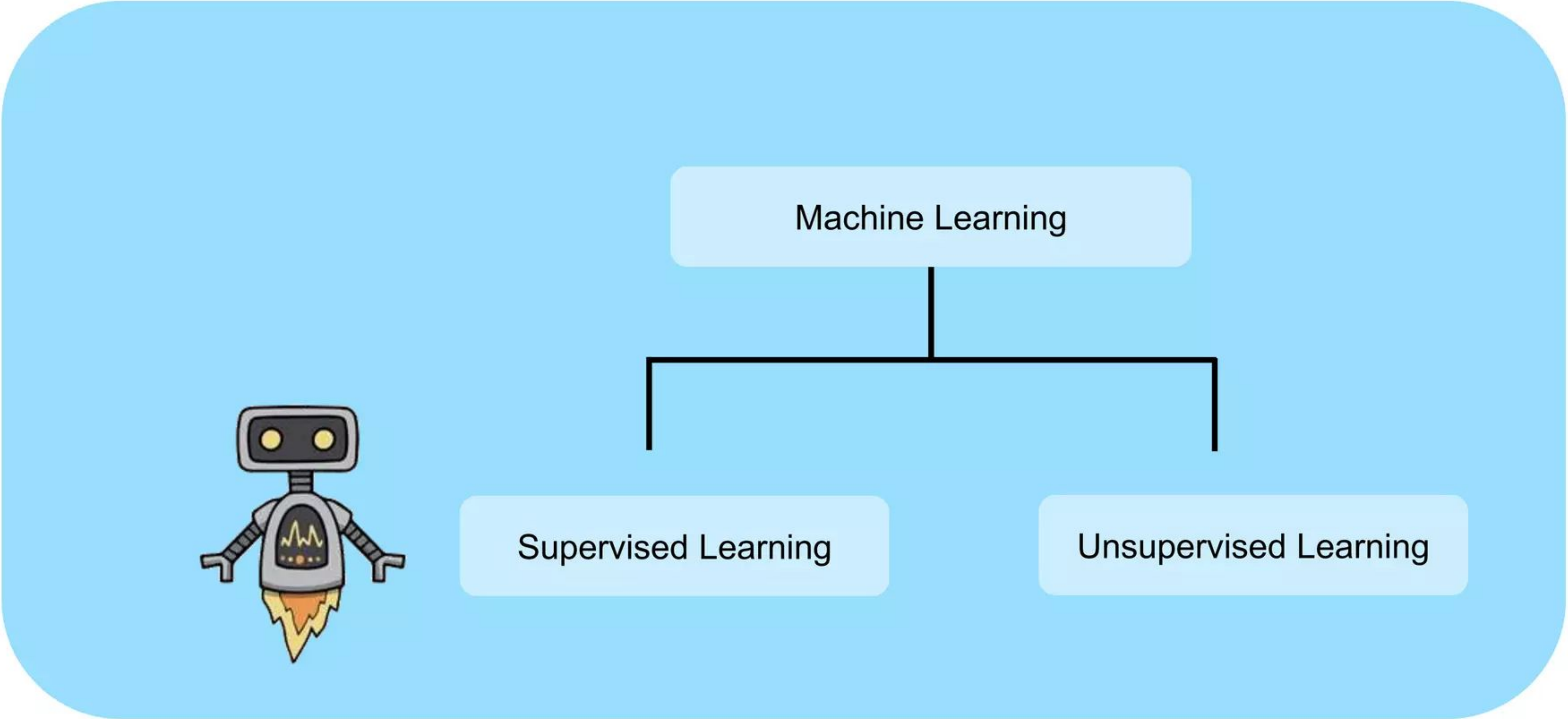
- AI in the medical field has the potential to improve patient care, increase efficiency, accelerate research, and enhance the overall healthcare experience.
- It complements the expertise of healthcare professionals, leading to more informed decision-making and better health outcomes for patients.

# Artificial Intelligence Vs. Machine Learning

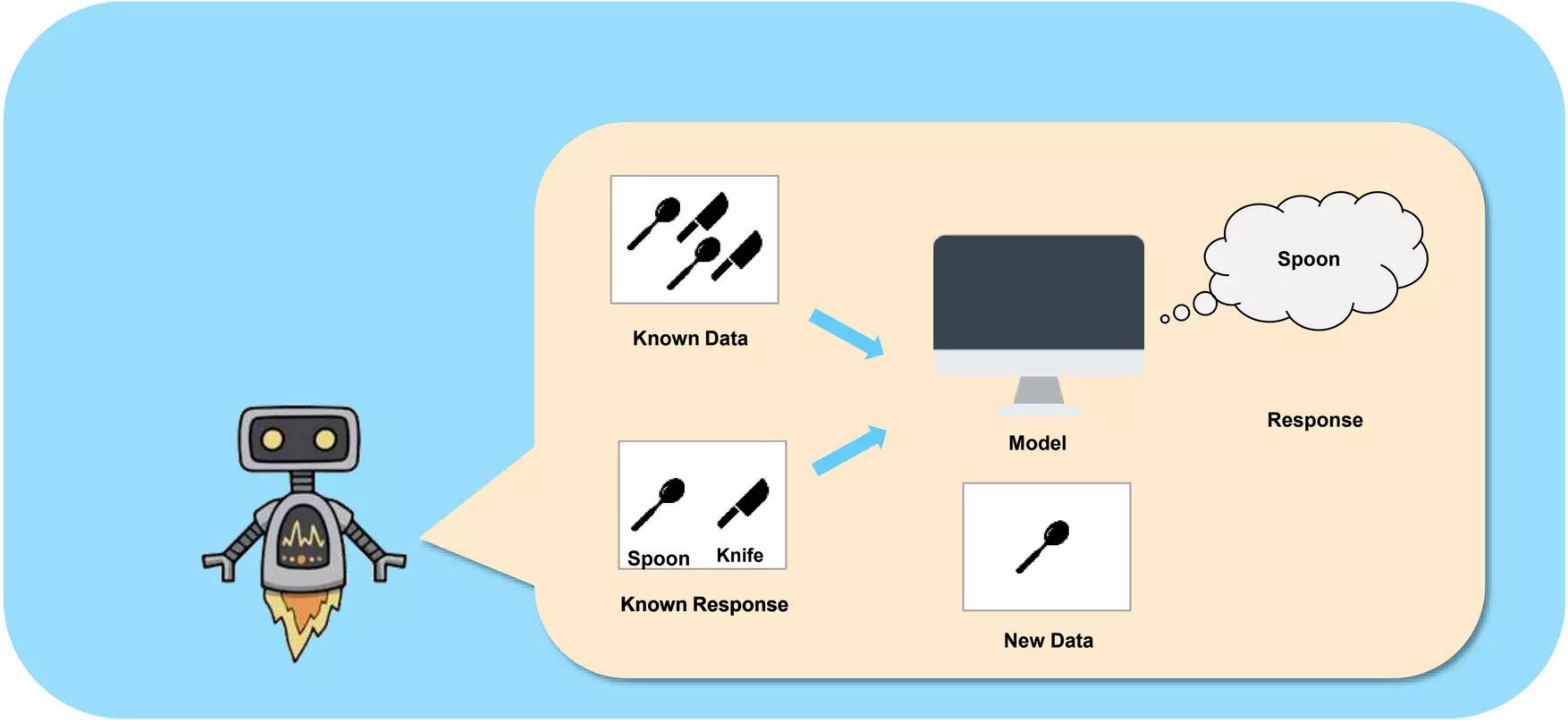
- **Machine Learning (ML)** A subset of AI, specifically involves the use of algorithms that allow machines to learn from and make predictions or decisions based on data.



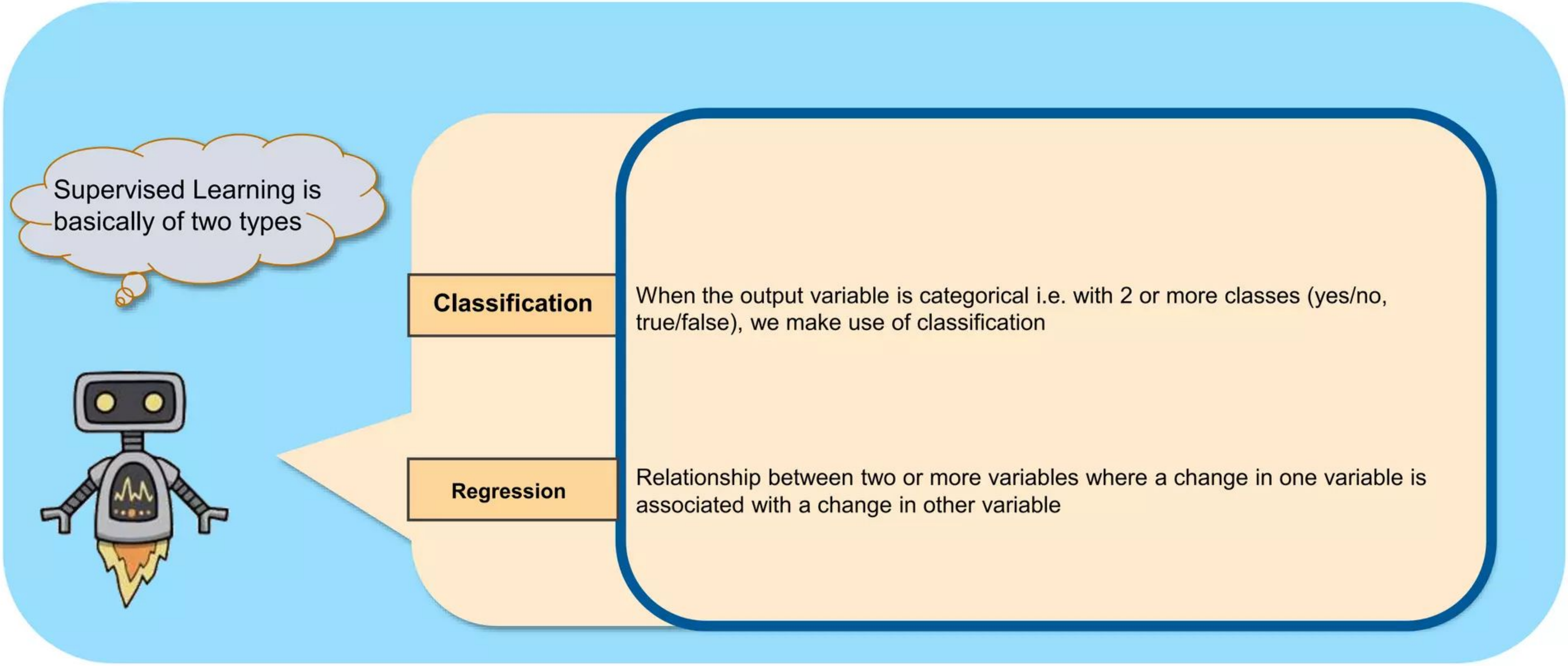
# Types of Machine Learning



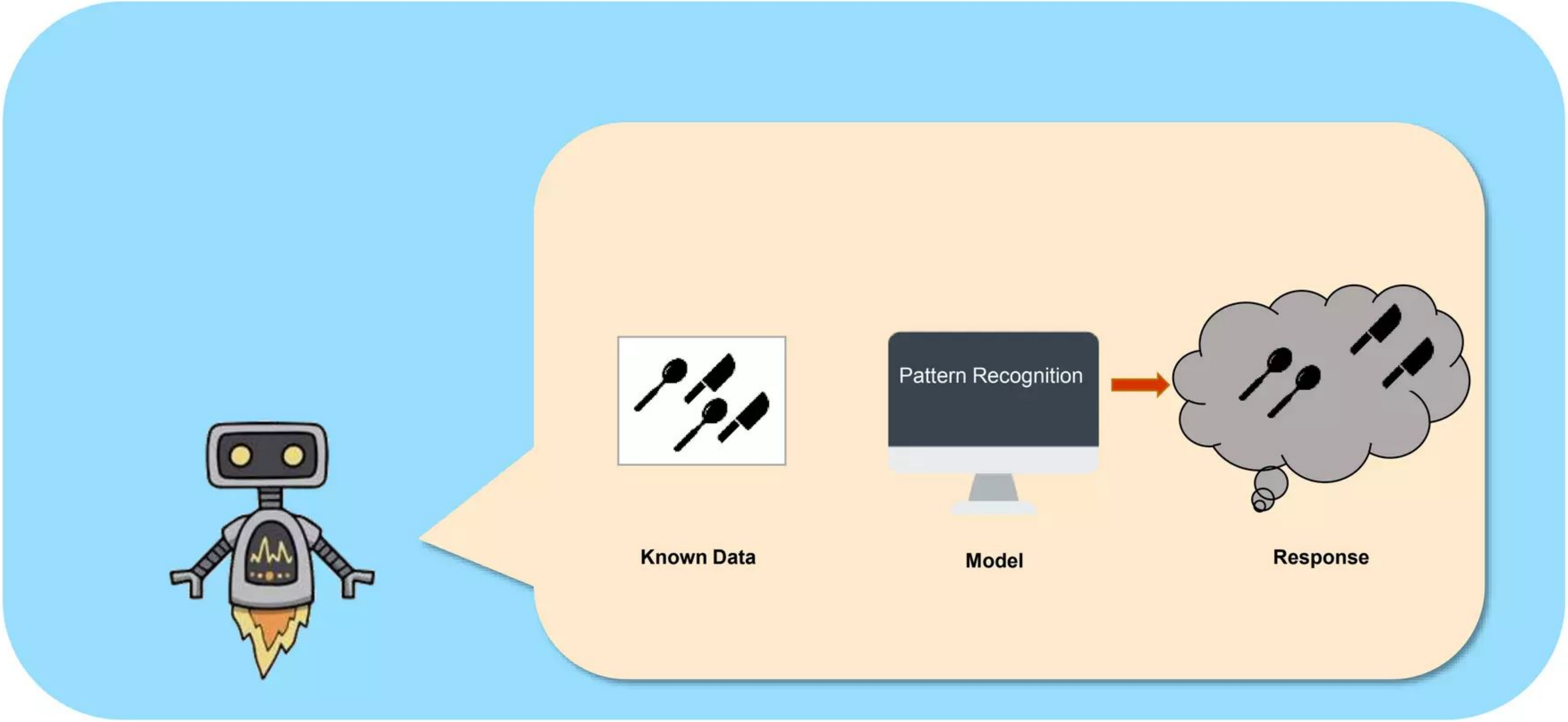
# Supervised Machine Learning



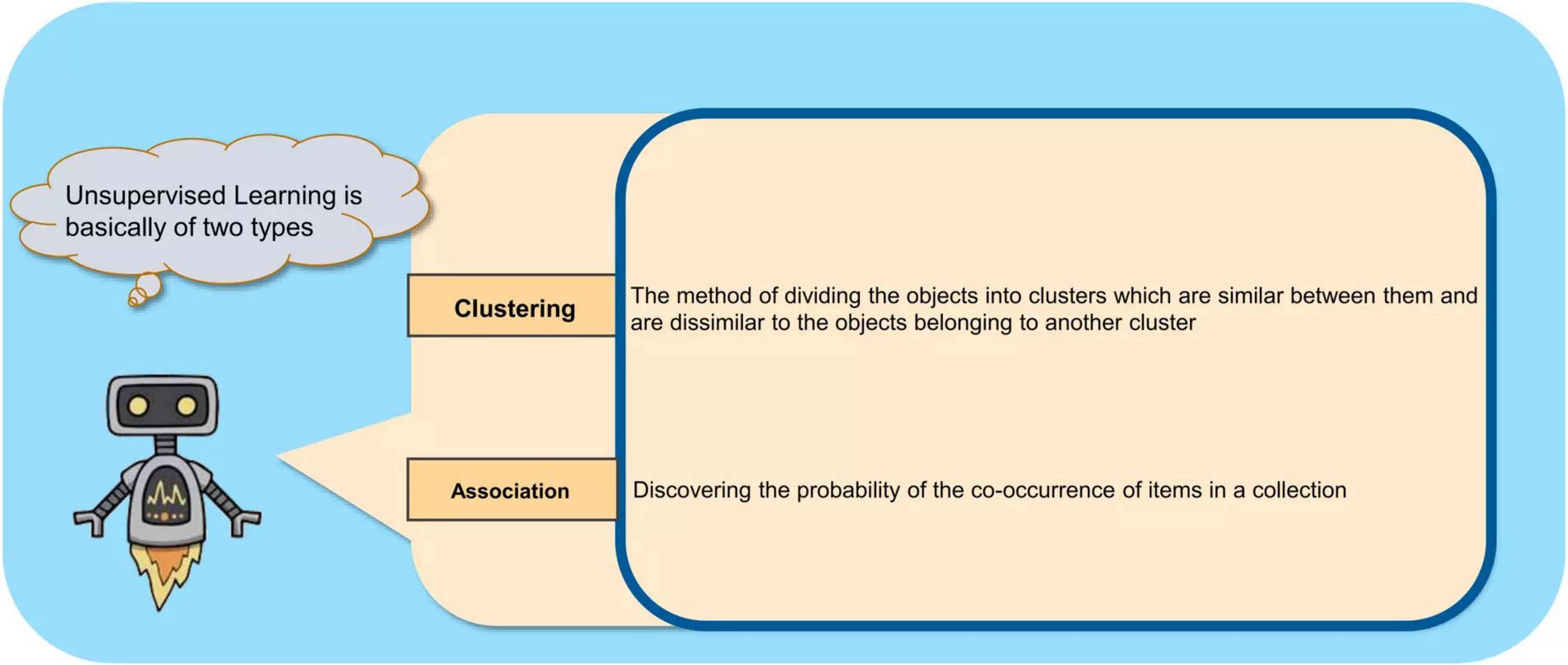
# Supervised Machine Learning



# Unsupervised Machine Learning



# Unsupervised Machine Learning



# Question 1

Can You Group the People in This Class?

- No predefined number of groups – create as many as you like.
- Use any criteria you think of to form the groups.
- Group sizes can vary - there are no restrictions.

# Question 2

Is she part of Group A or Group B?

What do you mean by Group A & B?

I will provide some examples of known people in these two groups, which may help ….

# Question 2

| sample | Feature 1 | Feature 2 | ... | Feature m | label |
|---|---|---|---|---|---|
| 1 | 1 | 0.6 | ... | 0.001 | A |
| 2 | 0 | 0.2 | ... | 0.1 | B |
| ... | ... | ... | ... | ... | ... |
| n | 1 | 0.4 | ... | 0.002 | A |
| Our sample | 1 | 0.5 | ... | 0.005 | ?????? |

n = number of samples m= number of features

# Question 2

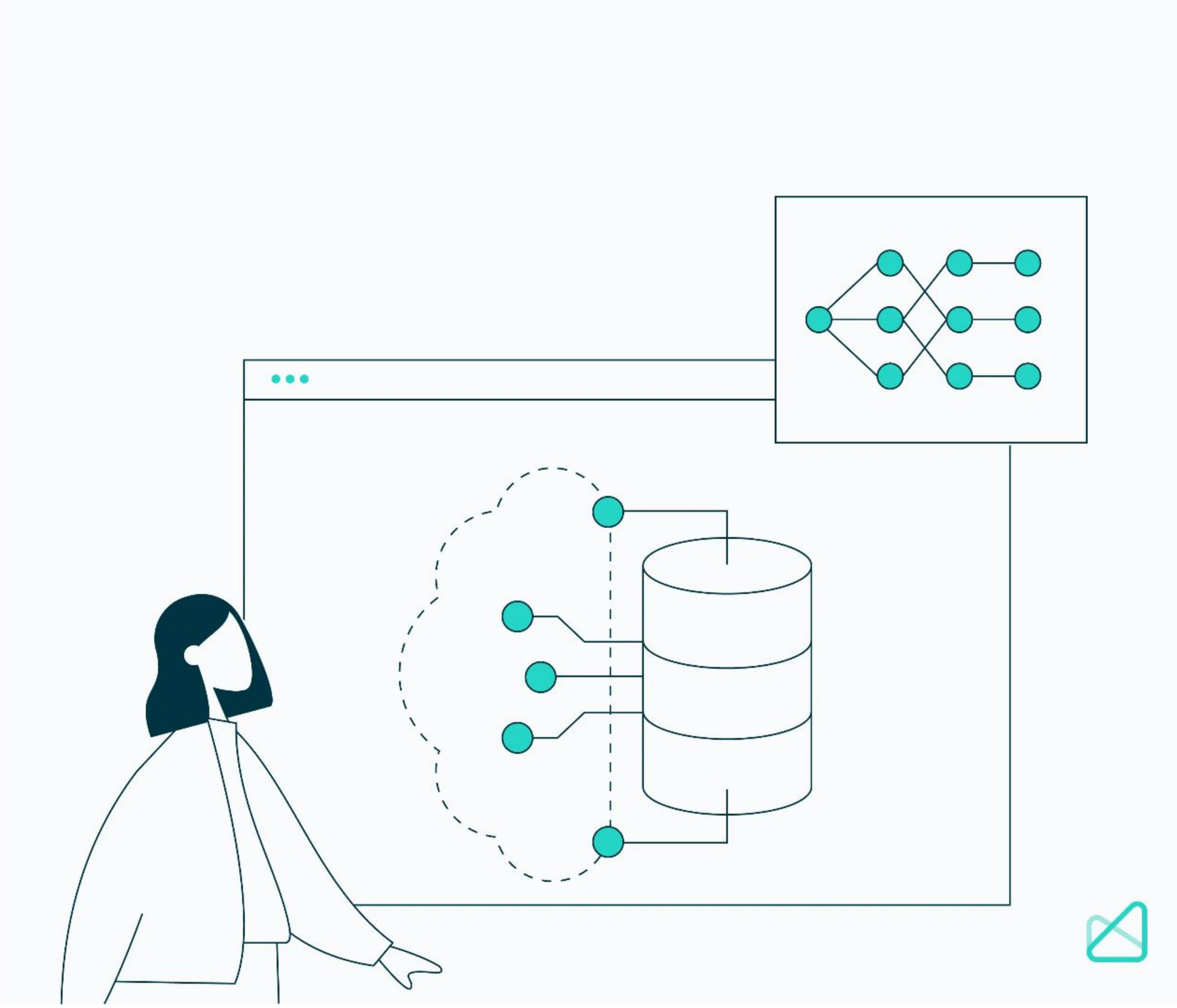
**Did the person attend the first session or not?**

| sample | Feature 1 | Feature 2 | ... | Feature m | label |  |
|---|---|---|---|---|---|---|
| 1 | 1 | 0.6 | ... | 0.001 | A | **Non-medical student** |
| 2 | 0 | 0.2 | ... | 0.1 | B | **Medical student** |
| ... | ... | ... | ... | ... | ... |  |
| n | 1 | 0.4 | ... | 0.002 | A |  |
| Our sample | 1 | 0.5 | ... | 0.005 | ?????? |  |

n = number of samples m= number of features

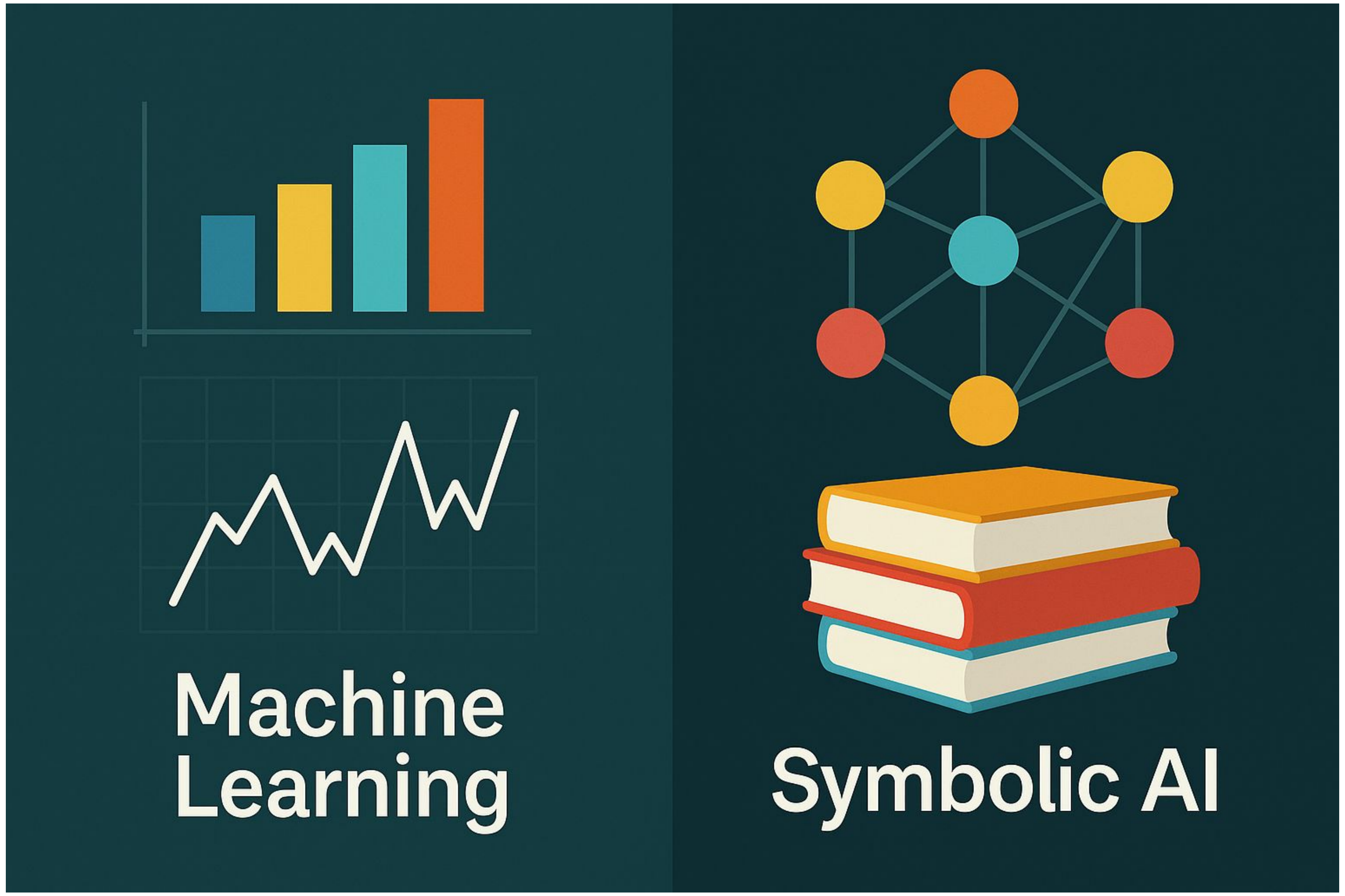
# When ML Is Not Enough

- Requires large, high-quality labeled datasets, often hard to obtain in healthcare.
- Struggles with rare conditions or limited data populations.
- Lacks explicit reasoning, models learn patterns but can't explain why.
- May produce biased results if training data isn't representative.
- Difficult to incorporate clinical guidelines or expert rules directly.



# From Patterns to Knowledge

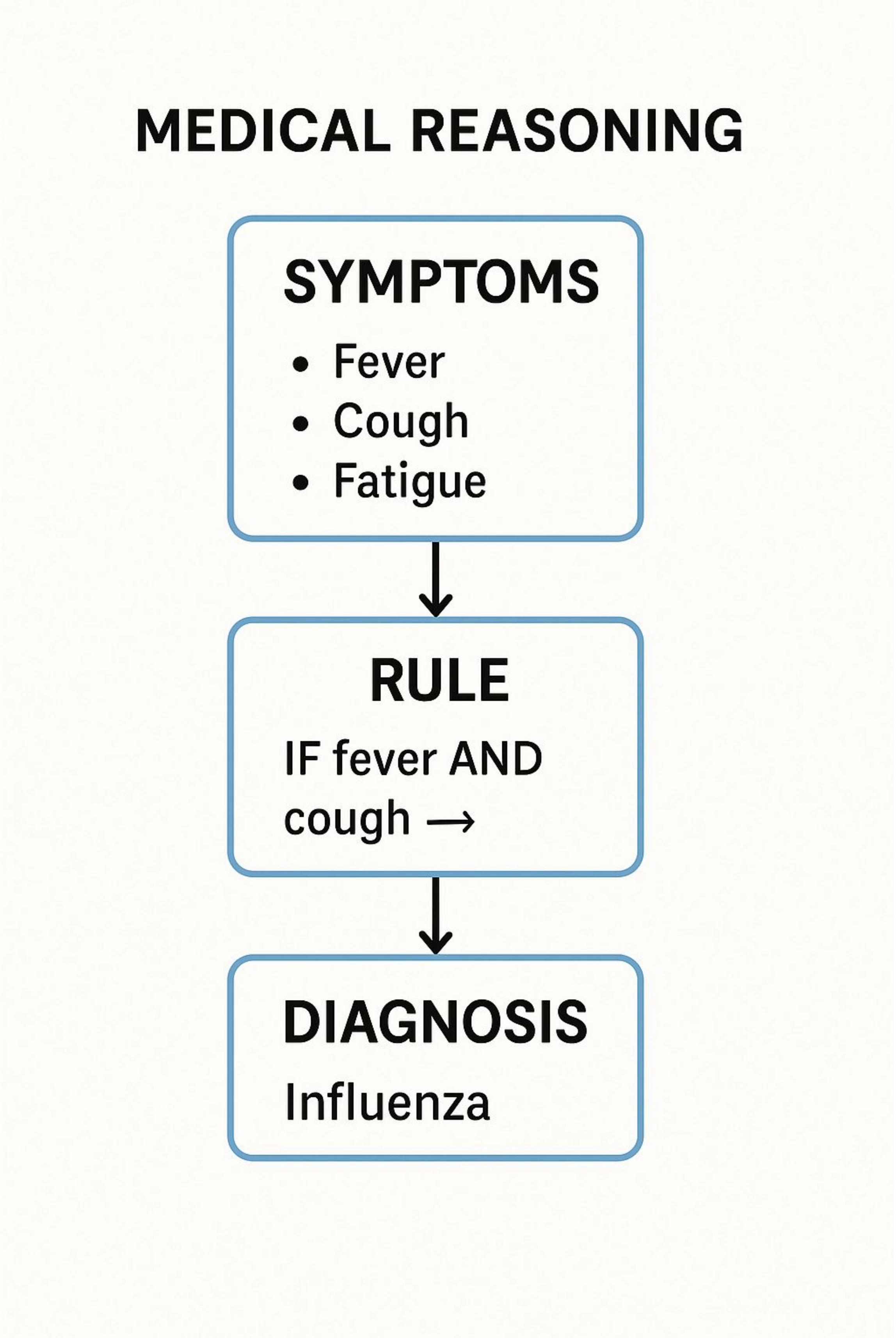
- Machine Learning discovers patterns in data, but medicine also depends on knowledge, logic, and reasoning.
- To combine the strengths of both, AI can also be built on medical knowledge, rules, and relationships, this is Symbolic or Knowledge-Based AI.



# Symbolic (Knowledge-based) AI in Medicine

- Relies on explicit rules, ontologies, and expert-defined relationships.
- Mimics the way clinicians reason: "If symptom A + test result B → consider diagnosis C."
- Useful when data is limited but medical knowledge is well established.
- Forms the basis for clinical decision support systems (CDSS) and diagnostic reasoning tools.

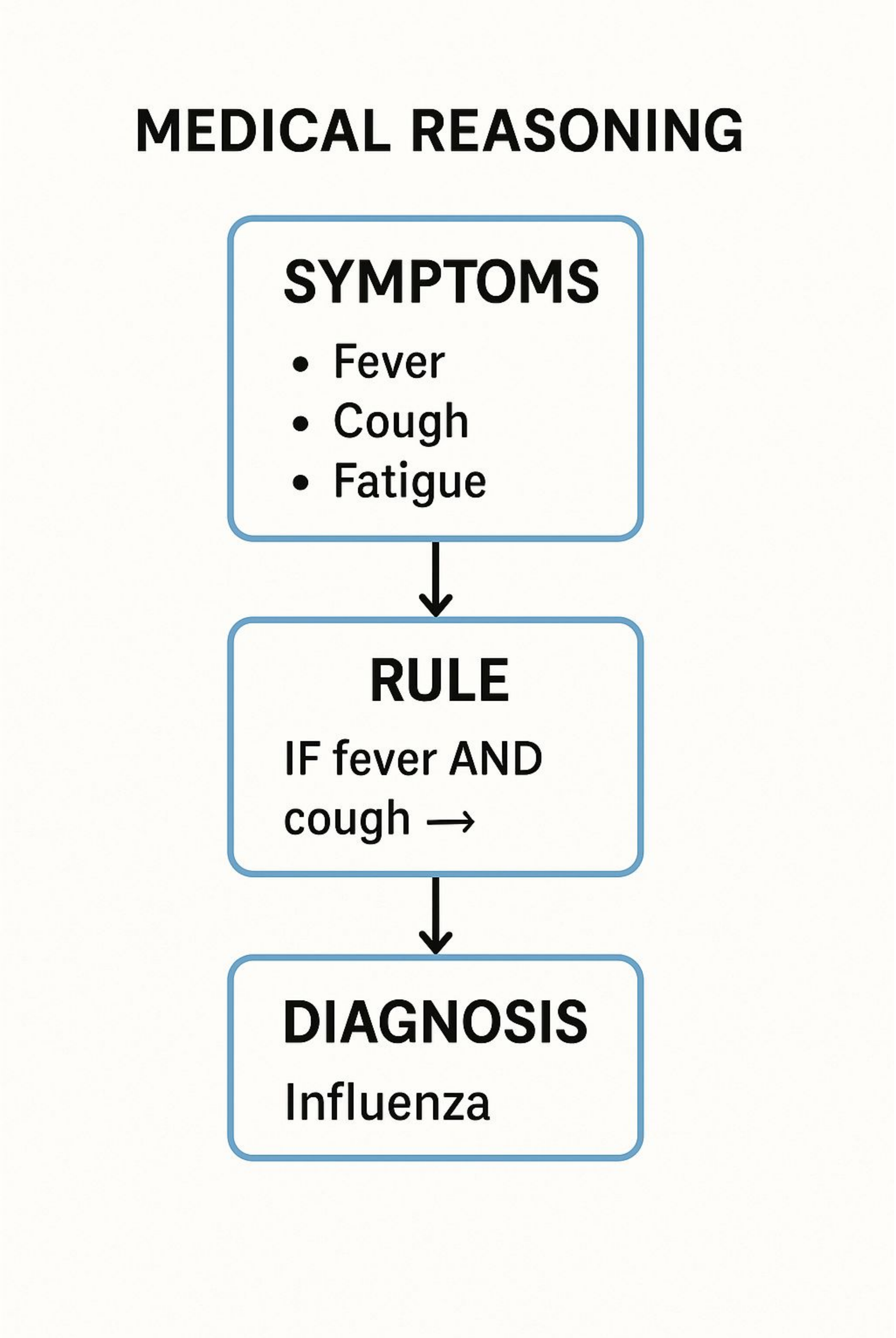
**Examples:**
- Drug–drug interaction checkers
- Rule-based triage systems (e.g., early diagnosis)



# Knowledge Graphs in Healthcare

- Knowledge Graphs represent relationships between medical entities (diseases, symptoms, treatments, genes, etc).
- Enable reasoning, discovery, and integration across diverse data sources.
- Help link structured knowledge (e.g., guidelines) with unstructured data (e.g., notes, imaging reports).
- Support explainable AI by showing why a certain connection or recommendation was made.

**Example:**
- Linking COVID-19 → symptoms → treatments → clinical trials.

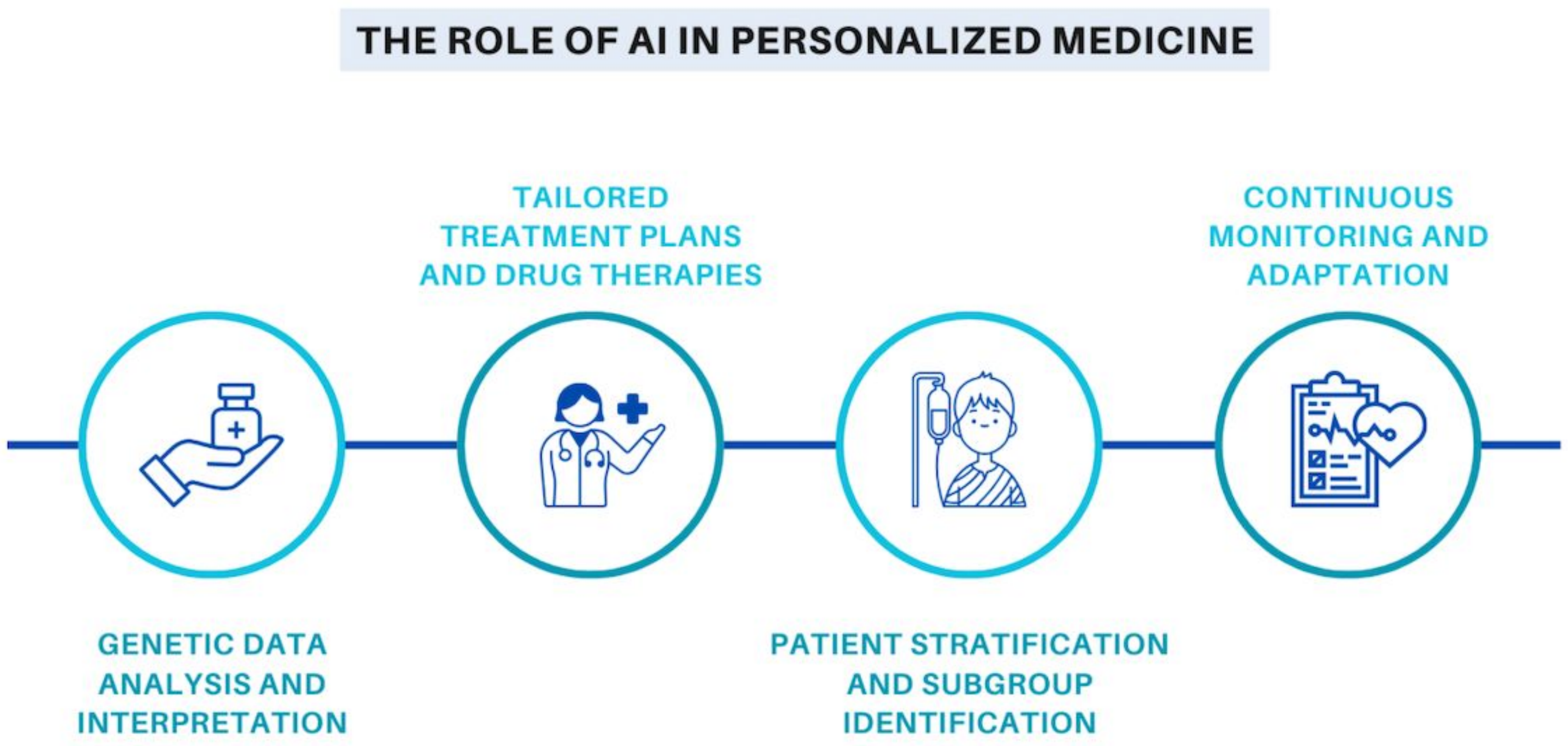


# How Can AI Help

| Application | Description |
|---|---|
| Diagnosis | AI algorithms can analyze medical images such as X-rays and 3D scans to detect abnormalities and make diagnoses |
| Predictive analytics | AI can analyze large amounts of data such as electronic medical records to predict the prospect of certain diseases and patient outcomes. |
| Personalized medicine | By analyzing a patient's medical and dental history and genetic makeup, AI can help create a personalized treatment plan |
| Drug discovery | AI can analyze compounds, predict their potential efficacy as drugs, and accelerate the drug discovery process |
| Clinical decision support | AI can provide physicians with real-time recommendations based on the latest evidence and guidelines |

# Personalized Medicine

- AI analyzes patient data including genetic information and medical history, to tailor treatment plans, medications, and therapies to individual patients, improving treatment outcomes.
- Algorithm designed takes input of patient, analyses, and gives medicine for disease.

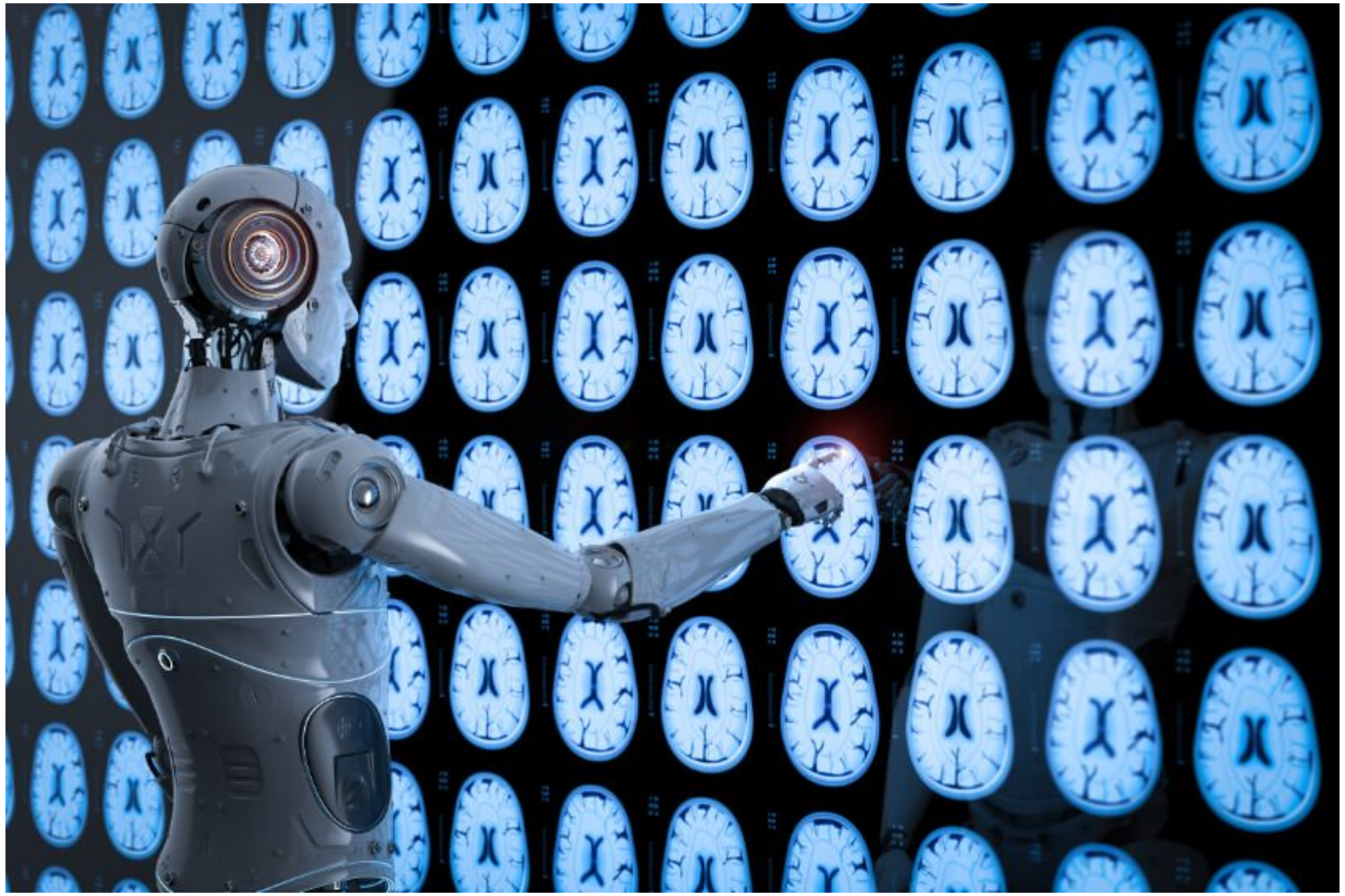


# Diagnosis and Treatment

- AI algorithms can analyze medical data and help detect diseases and abnormalities quickly, aiding in early and accurate disease diagnosis.

# Medical Imaging

- AI algorithms can analyze medical images (CT scans, MRI, X-rays), helping to detect diseases and anomalies with high accuracy. This speeds up diagnosis and reduces the risk of human error.



# Robotic Surgery

- AI-assisted robotic surgery systems enhance the precision and capabilities of surgeons,
- leading to better surgical outcomes and reduced invasiveness.

# Healthcare Chatbots

- AI-powered chatbots and virtual assistants provide support for patients, answering medical questions, scheduling appointments, and providing health information.

- AI-powered devices enable continuous remote patient monitoring, allowing healthcare providers to track patients' vital signs and health conditions in real-time, improving patient care and reducing hospital readmissions.

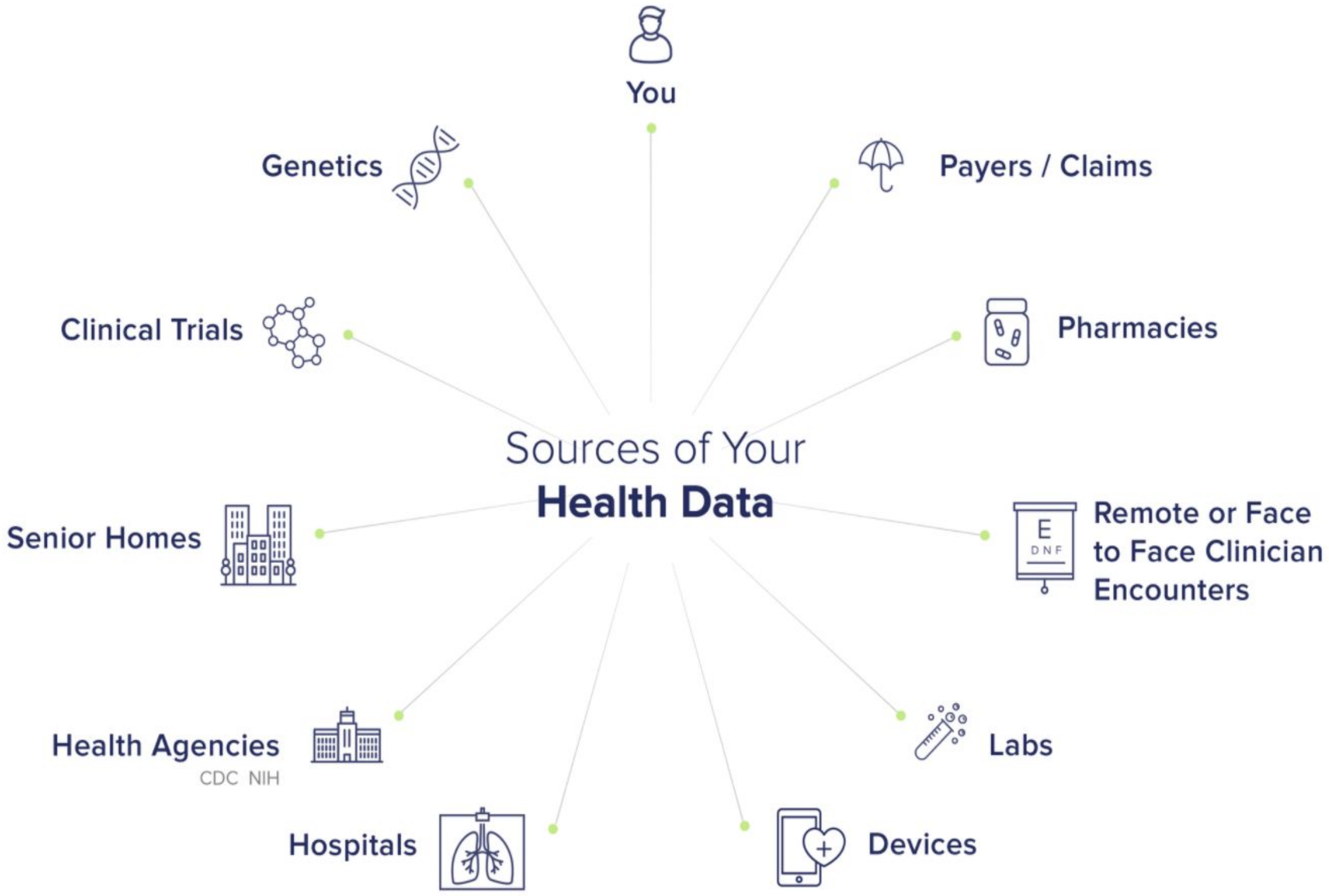
# Genomic Medicine and Gene Mapping

- AI aids in the analysis of genetic data, helping identify genetic markers for diseases and potential treatment options.

# Drug Discovery

- AI can accelerate drug development by analyzing vast datasets to identify potential drug candidates and predict their effectiveness.

# Where Does Health Data Come From



# Data Sources

**1- Primary data source**
Team collects data to answer specific question.
**PROS:** Able to control quality of data collection and customize collection based on the features needed.
**CONS:** Very time consuming, more expensive, lots of ethics paperwork, extensive pre-processing steps required.

**2- Secondary data source**
Team requests data from a pre-existing database or study
**PROS:** Quick and cheap, able to gather large number of datapoints
**CONS**: May not have access to all the features you would want

# Some Related Technical Terms

**1- Data Verification**
The process of examining various types of data for consistency and accuracy after data migration

**2- Data Credibility**
How complete and accurate a data set is

**3- Data Cleaning**
The process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset.

**4- Data Privacy**
The ability of a person to determine for themselves when, how, and to what extent personal information about them is shared with or communicated to others.

**5- Data Ethics**
The field of study that addresses questions about the appropriate use of data

**6- Data Access**
Generic term referring to a process which has both an IT-specific meaning and other connotations involving access rights in a broader legal and/or political sense.

# 5 V's of Big Data

Volume refers to the amount of data that exists.

Velocity refers to how quickly data is generated and how quickly that data moves.

Variety refers to diversity of data types.

Veracity refers to the quality and accuracy of data.

Value refers to the quantity that the data contains.

# Introduction to Health Data

**Structured**

Standardized

Easily transferable between health information systems

Ex: Age, gender, etc.

**Unstructured**

Not standardized

Significant pre-processing required prior to use

Ex: Radiographs (X-Rays and CBCT), scans, charting notes, etc.

# Introduction to Health Data

### Quantitative

Uses numbers

Easily transferable between health information systems

Ex: Age, weight, probing depth, etc.

### Qualitative

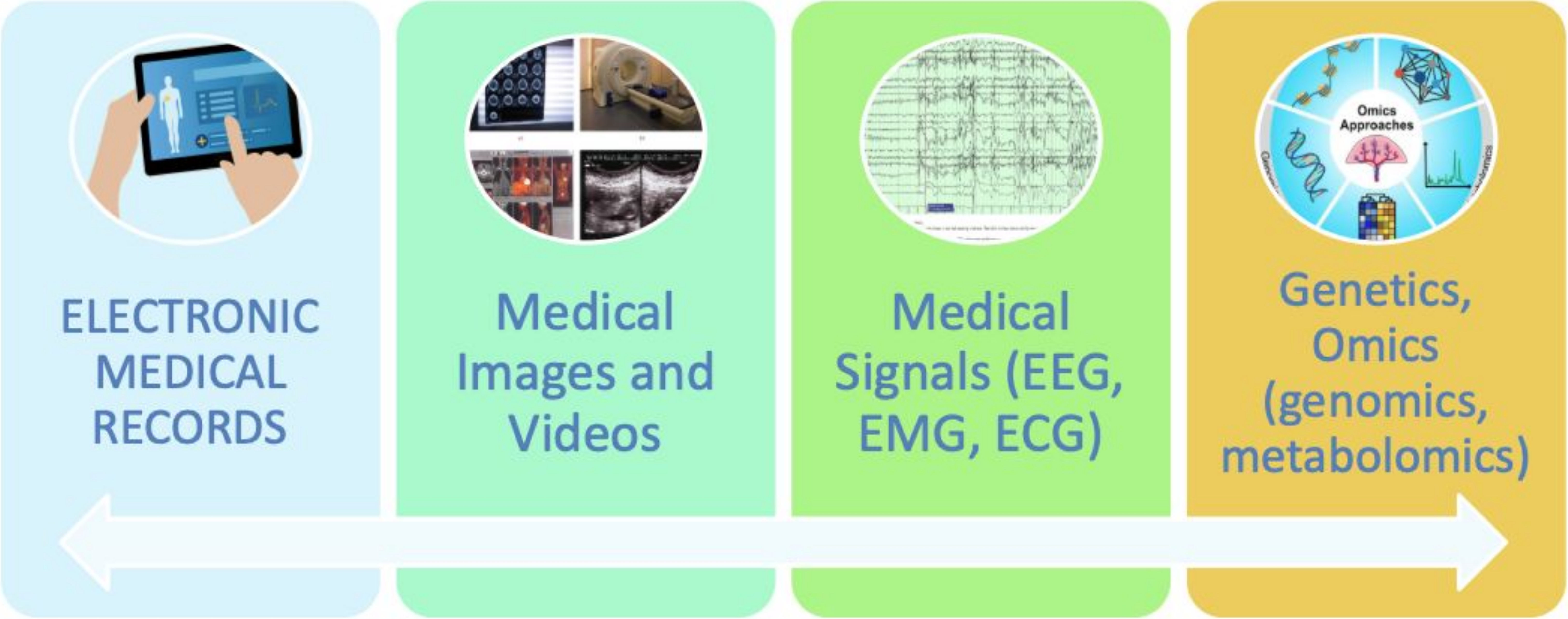
Non-numerical

Usually observational

Transferability to different health systems depends on the data

"1730h: Mr. Smith presented with sharp headache"

# Example of Different Types of Data in Medicine

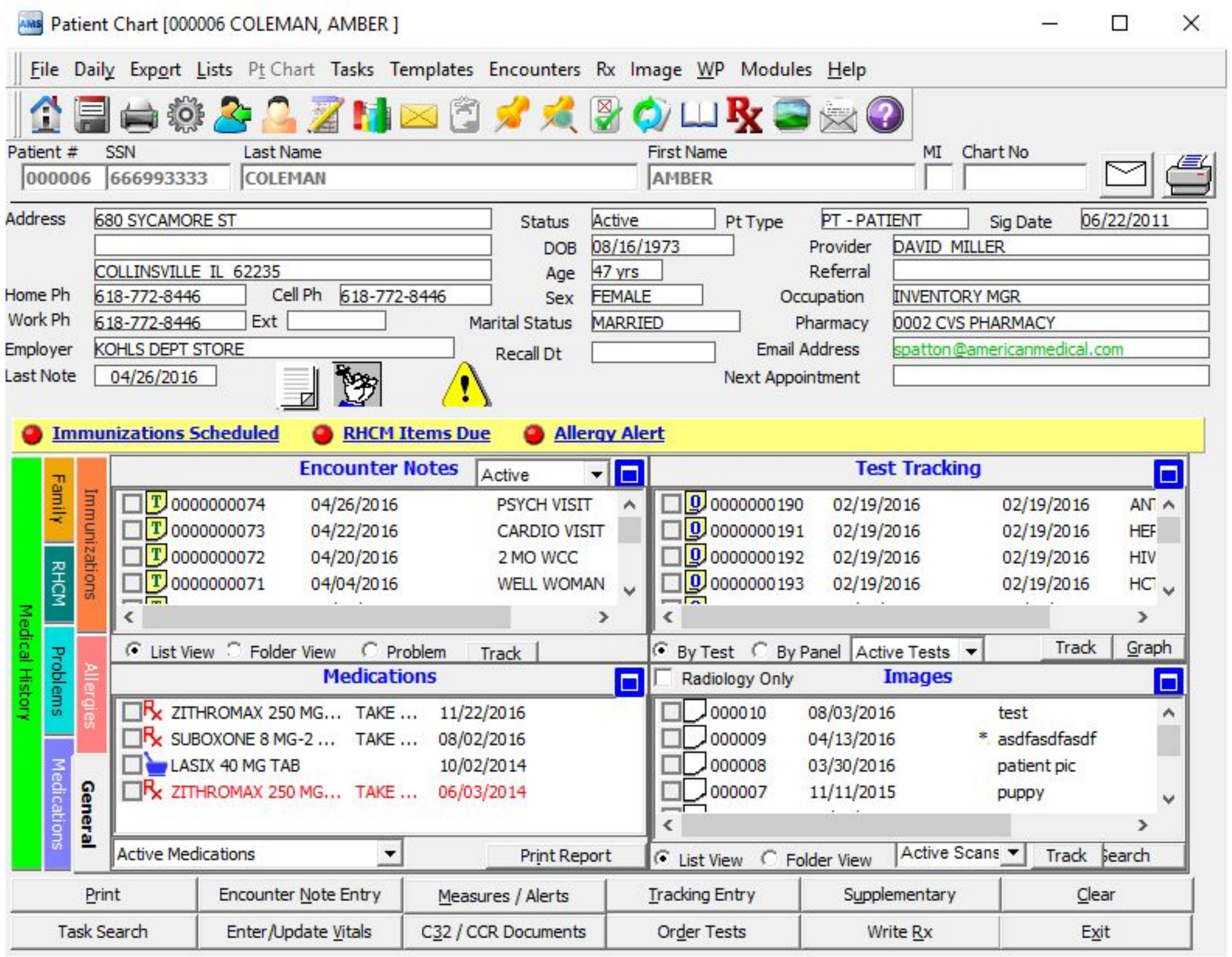


# Electronic Medical Records

# EMR Introduction

**Key idea:** centralized and standardized health records

Use of computers in medical practices in the United States has increased gradually over the past decades. In 1983–84, approximately 19.7% of family physicians reported using a computer in their office, with around 14.0% owning or leasing one.



# What Does EMR Data Include

# Advantages of EMR Data

- Both broad and deep
    - Contains thorough detail about an individual
    - Does this for (often) millions of patients!

- Deep learning often works best with this size of data

- This data can be used to learn both about individuals, populations, healthcare operations management, and more

- The data itself is a byproduct, we aren’t spending extra money to generate it for research purposes

# Drawbacks of EMR Data

- Completeness/quality/integrity issues
- Incredibly sensitive information
- Sheer scale
- Many different data types
- Longitudinal complexity

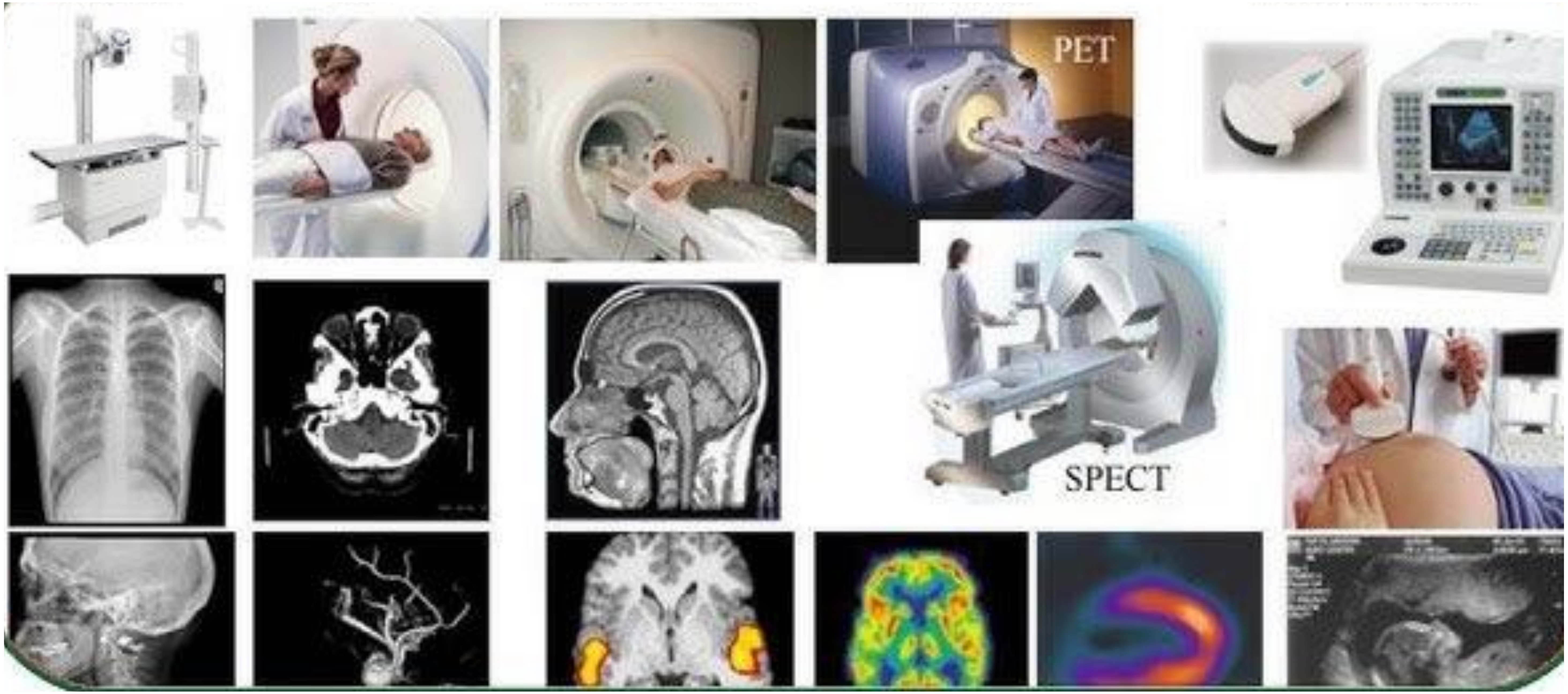
# Medical Imaging

# Introduction

- With recent advancements in medical imaging technology, clinicians can now capture highly detailed anatomical and functional information at both micro and macro levels, greatly improving disease detection, diagnosis, and treatment planning.

- **Medical Imaging:** Refers to the use of advanced imaging modalities such as X-ray, CT, MRI, ultrasound, and PET to visualize internal body structures for diagnosis, treatment planning, and monitoring of medical conditions.

# Types of Medical Imaging



# Advantages of Medical Imaging Data

**Higher Diagnostic Accuracy**
Digital imaging enhances visualization of anatomical structures, supporting early detection of diseases and abnormalities.

**Enhanced Treatment Planning**
3D imaging modalities such as CT and MRI provide detailed anatomical information for precise surgical and interventional planning.

**Easier Data Retrieval & Sharing**
Cloud-based PACS and EMR systems enable instant access to previous images and seamless sharing among healthcare providers.

**Lower Radiation Exposure**
Digital imaging technologies reduce radiation dose compared to conventional film-based methods.

**AI-Assisted Analysis**
Machine learning algorithms can automate image interpretation tasks such as lesion detection, organ segmentation, and disease classification.

# Disadvantages of Medical Imaging Data

**Large File Sizes**
High-resolution CT, MRI, and PET images require substantial storage space and may slow retrieval times.

**High Initial Investment**
Advanced imaging equipment such as MRI, CT, and PET scanners are expensive to purchase and maintain.

**Training & Software Dependency**
Accurate interpretation and workflow efficiency depend on specialized training and advanced imaging software.

**Image Artifacts & Errors**
Motion, metal implants, or technical issues can produce artifacts that affect image quality and diagnostic accuracy.

**Privacy & Security Risks**
Cloud-based image storage and PACS systems may be vulnerable to data breaches if not properly encrypted or managed.

# Introduction

- Charting and clinical documentation are essential for recording patient history, diagnosis, and treatment plans in modern healthcare.

- Electronic Medical Record (EMR) systems have replaced traditional paper charts, improving data structure, accessibility, and clinical efficiency.

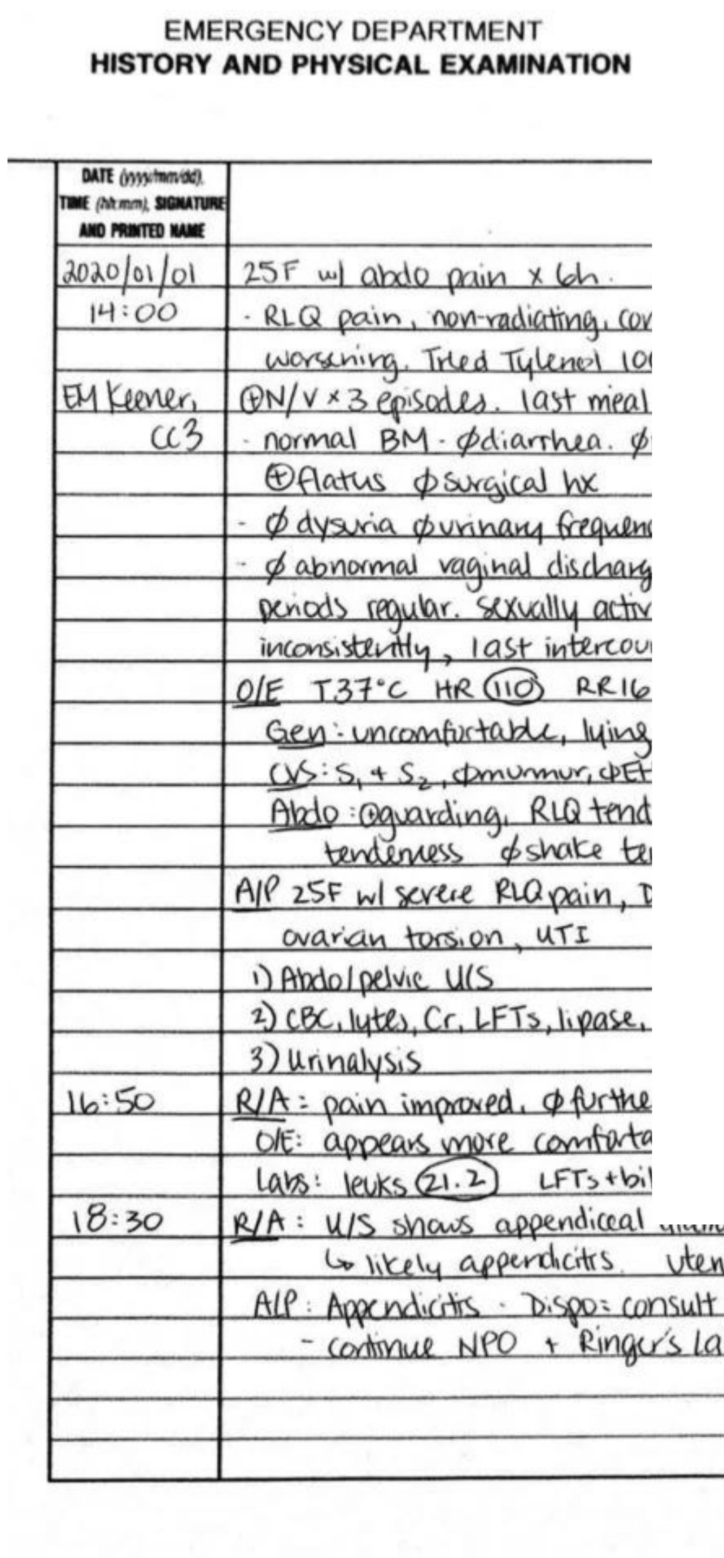
- These systems capture vital signs, diagnostic results, medication lists, and progress notes, while clinical documentation details diagnoses, interventions, and follow-up care.

- Advanced features such as speech-to-text entry and AI-assisted documentation enhance accuracy, reduce clinician workload, and streamline clinical workflows.

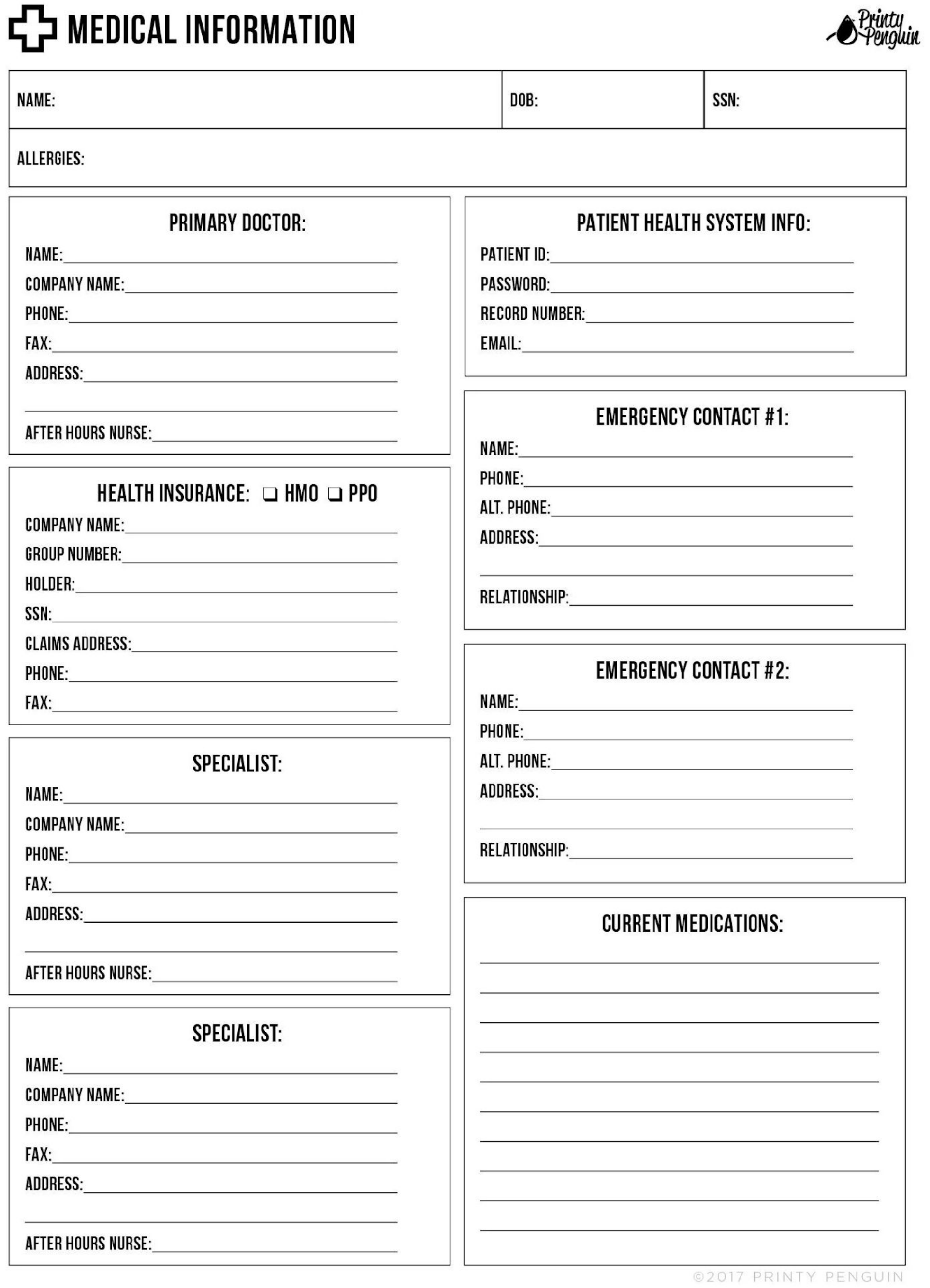
# Charting and Clinical Notes

# Types of Charting and Clinical Notes

Includes vital sign charting, medication records, progress notes, diagnostic test reports, imaging annotations, and voice-to-text transcriptions each providing detailed insights into different aspects of patient care and clinical management.


EMERGENCY DEPARTMENT
HISTORY AND PHYSICAL EXAMINATION

| DATE (yyyy/mm/dd) TIME (hh:mm), SIGNATURE AND PRINTED NAME | |
|---|---|
| 2020/01/01 14:00 | 25F w/ abdo pain x 6h. |
| | - RLQ pain, non-radiating, co |
| | worsening. Tried Tylenol 10 |
| EM Keener, CC3 | ⊕N/V x3 episodes. last meal |
| | - normal BM - ø diarrhea. ø |
| | ⊕flatus ø surgical hx |
| | - ø dysuria ø urinary frequen |
| | - ø abnormal vaginal discharg |
| | periods regular. sexually activ |
| | inconsistently, last intercou |
| | O/E T37°C HR 110 RR16 |
| | Gen: uncomfortable, lying |
| | CVS: $S_1$ + $S_2$, ø murmur, ø E |
| | Abdo: ø guarding, RLQ tend |
| | tenderness ø shake te |
| | A/P 25F w/ severe RLQ pain, D |
| | ovarian torsion, UTI |
| | 1) Abdo/pelvic U/S |
| | 2) CBC, lytes, Cr, LFTs, lipase, |
| | 3) Urinalysis |
| 16:50 | R/A: pain improved. ø furthe |
| | O/E: appears more comforta |
| | Labs: leuks 21.2 LFTs + bil |
| 18:30 | R/A: U/S shows appendiceal |
| | ↳ likely appendicitis. Uterus/ovaries/fallopian tubes unremarkable |
| | A/P: Appendicitis - Dispo: consult General Surgery for admission/surgery |
| | - continue NPO + Ringer's lactate 100cc/h maintenance |


MEDICAL INFORMATION
Printy Penguin

NAME: DOB: SSN:
ALLERGIES:

PRIMARY DOCTOR:
NAME:
COMPANY NAME:
PHONE:
FAX:
ADDRESS:
AFTER HOURS NURSE:

PATIENT HEALTH SYSTEM INFO:
PATIENT ID:
PASSWORD:
RECORD NUMBER:
EMAIL:

HEALTH INSURANCE: ❑ HMO ❑ PPO
COMPANY NAME:
GROUP NUMBER:
HOLDER:
SSN:
CLAIMS ADDRESS:
PHONE:
FAX:

EMERGENCY CONTACT #1:
NAME:
PHONE:
ALT. PHONE:
ADDRESS:
RELATIONSHIP:

SPECIALIST:
NAME:
COMPANY NAME:
PHONE:
FAX:
ADDRESS:
AFTER HOURS NURSE:

EMERGENCY CONTACT #2:
NAME:
PHONE:
ALT. PHONE:
ADDRESS:
RELATIONSHIP:

SPECIALIST:
NAME:
COMPANY NAME:
PHONE:
FAX:
ADDRESS:
AFTER HOURS NURSE:

CURRENT MEDICATIONS:

# Advantages of Charting and Clinical Notes

**Standardized Documentation**
Digital charting eliminates illegible handwritten notes and ensures consistency.

**Improved Treatment Tracking**
Provides an organized history of treatments and periodontal status.

**Easier Integration with AI & Decision Support**
AI-assisted notes can flag missing data and suggest treatments.

**Faster Input Methods**
Speech-to-text technology can speed up note-taking.

**Secure & Remote Access**
Cloud-based records allow dentists to review patient charts from multiple locations.

# Disadvantages of Charting and Clinical Notes

**Training and Adaptation**
Transitioning from paper-based to digital notes may require training.

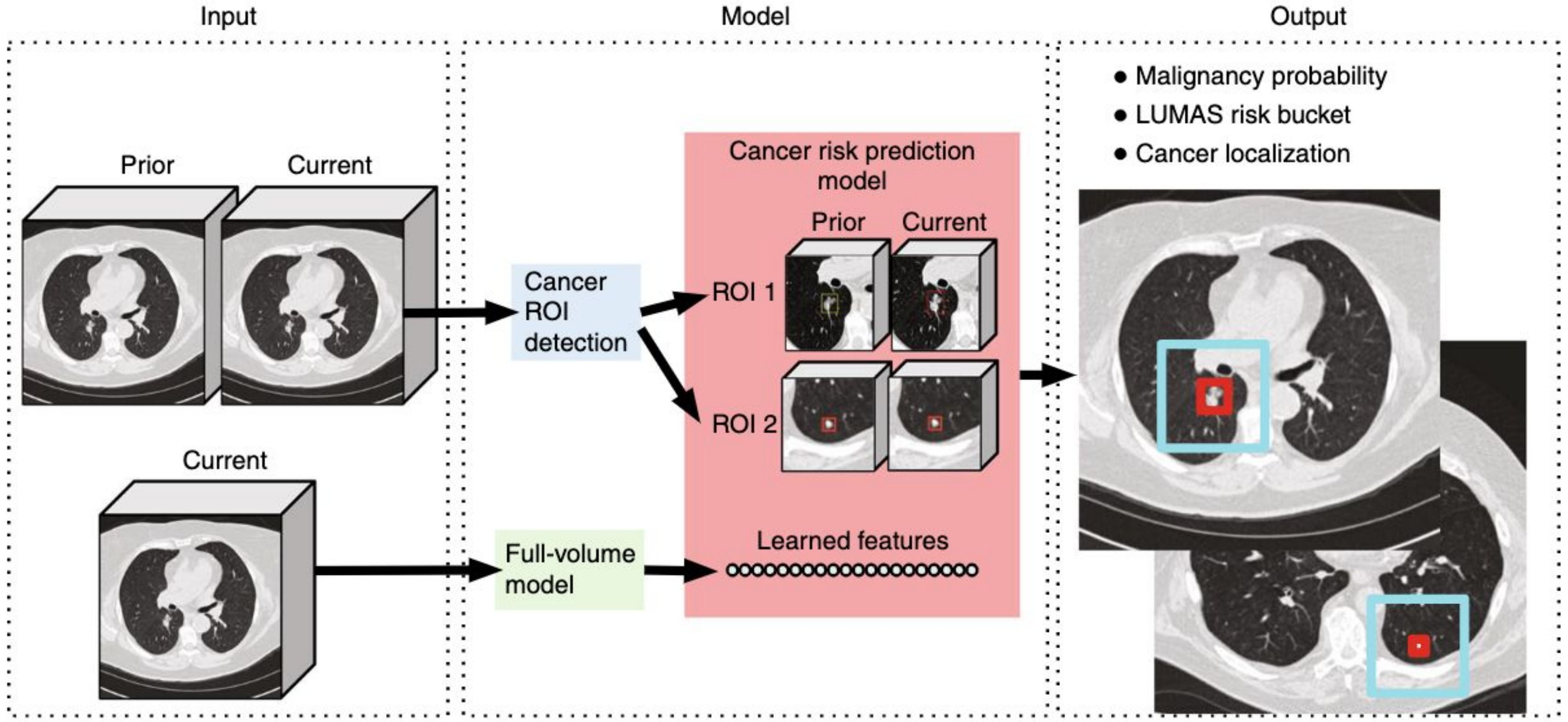
**Speech Recognition Errors**
Voice-to-text systems may misinterpret dental terminology, leading to inaccurate records.

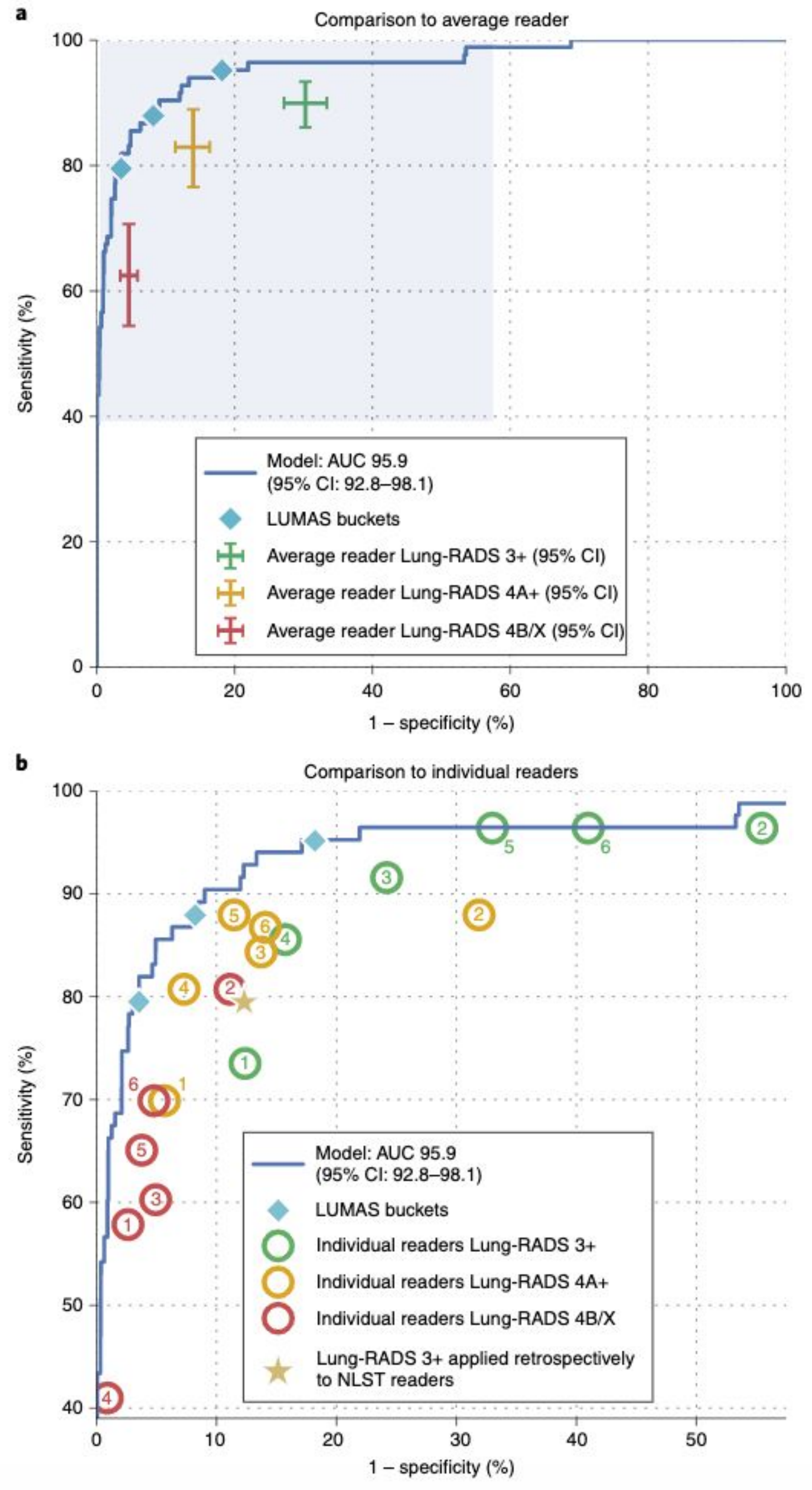
**Ethical Issues**
Digital records are susceptible to hacking if not properly encrypted.

**Incomplete or Inaccurate Entries**
Errors in manual entry or auto-generated notes can impact diagnosis and treatment.

# Screening

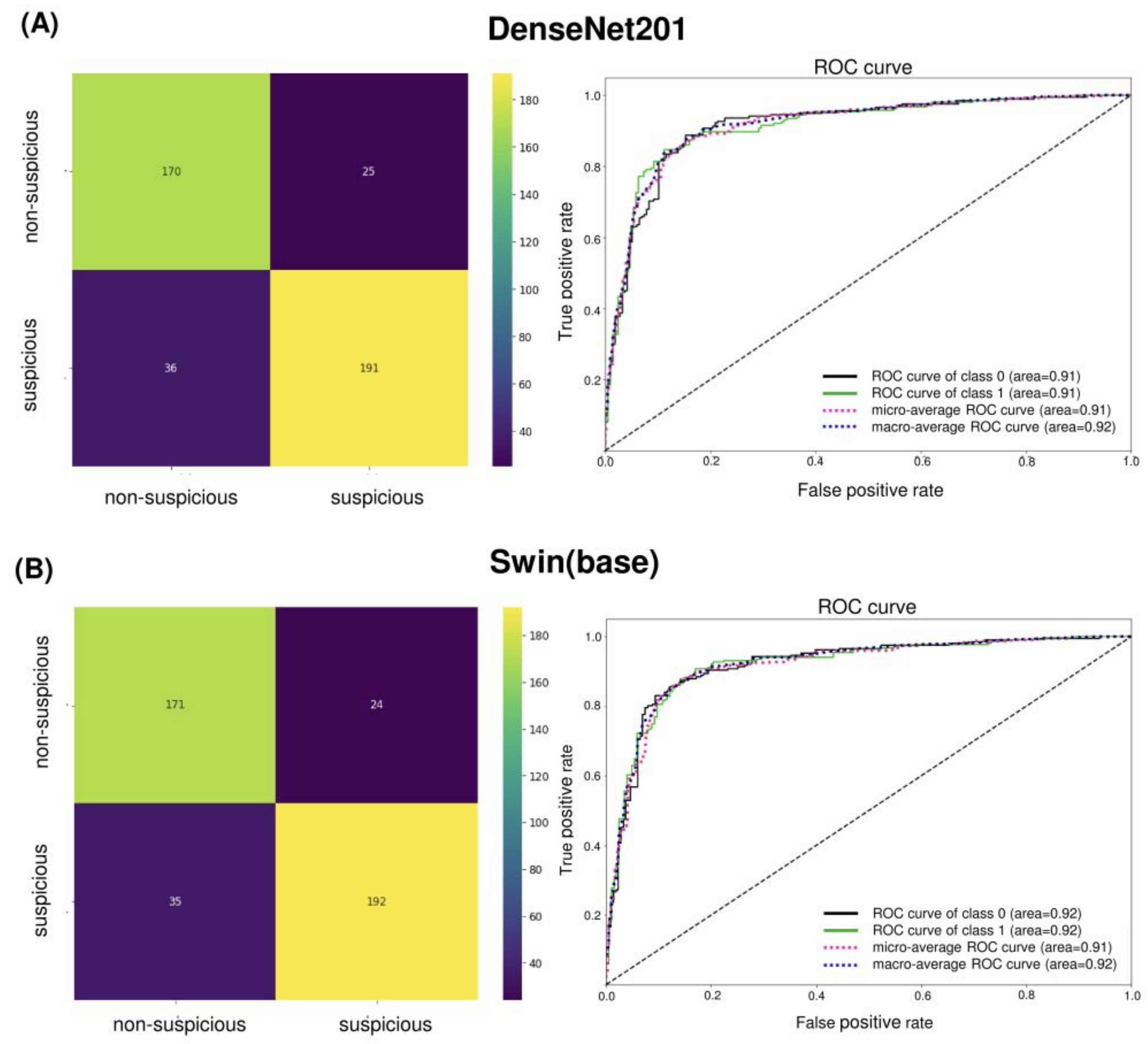
End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography (chest CT) (Ardila et al, 2019)

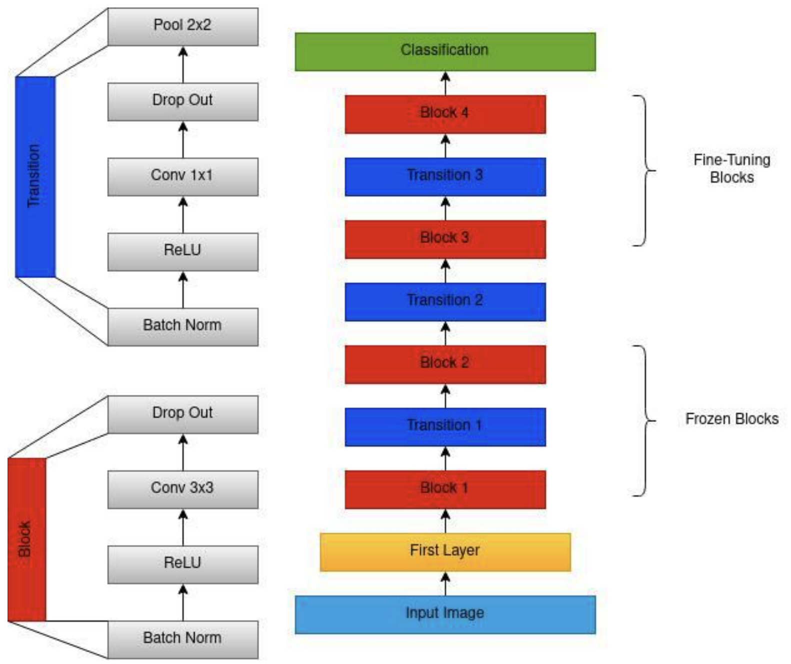


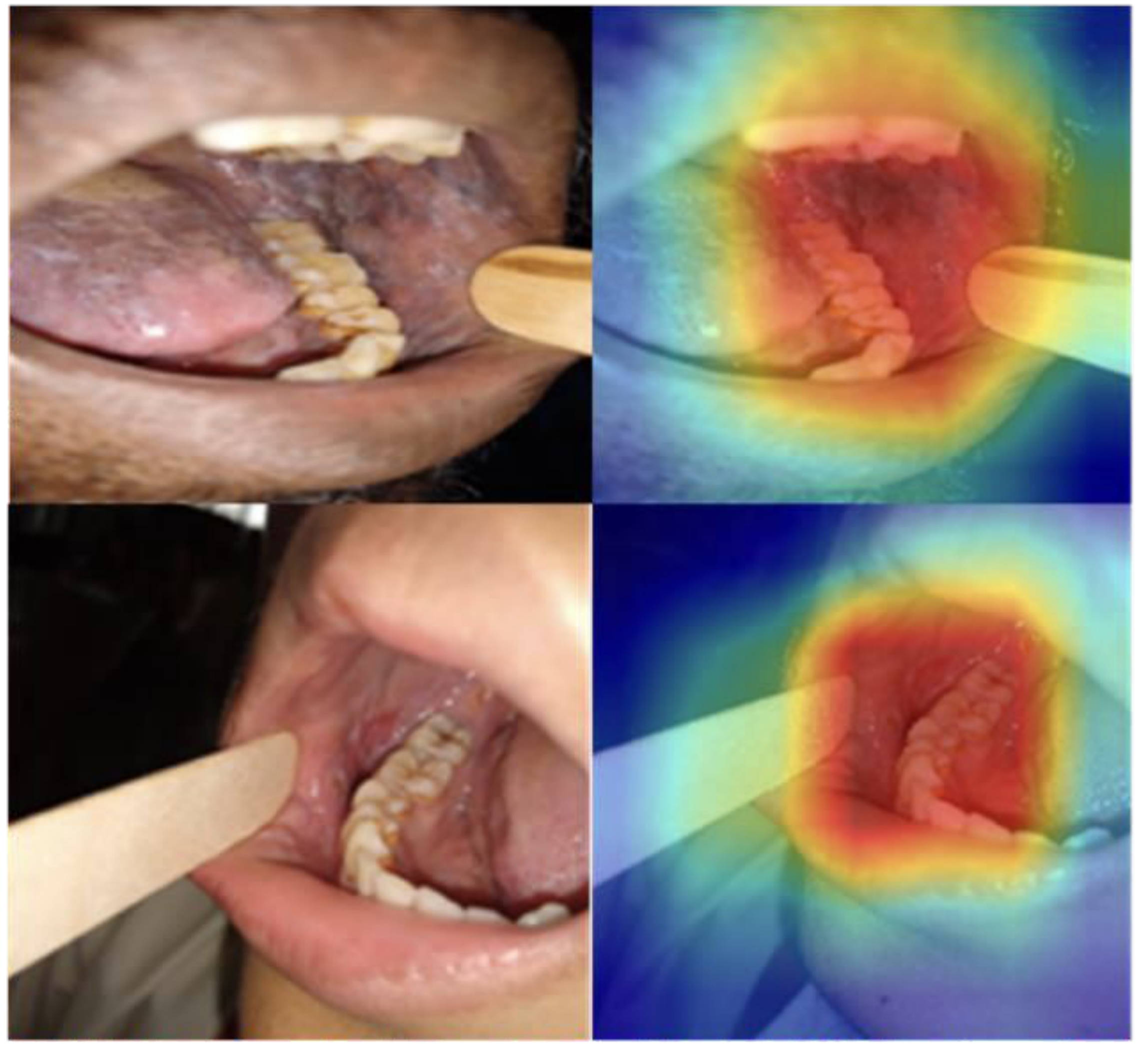



# Screening

## AI-Assisted Screening of Oral Potentially Malignant Disorders Using Smartphone-Based Photographic Images (Talwar et al., 2023)

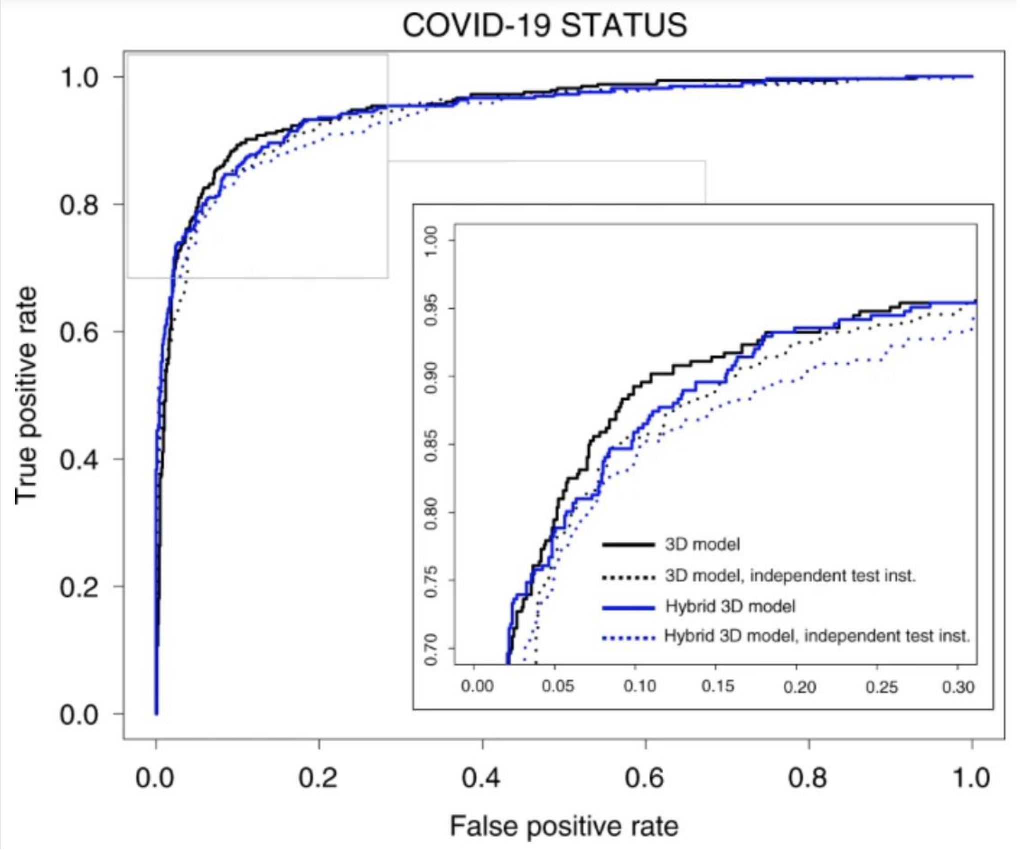


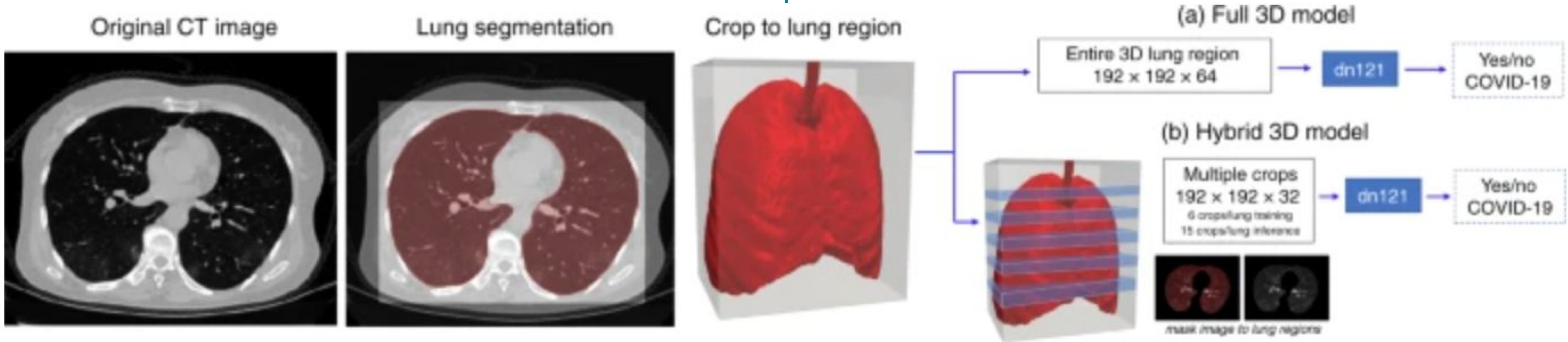



# Diagnosis

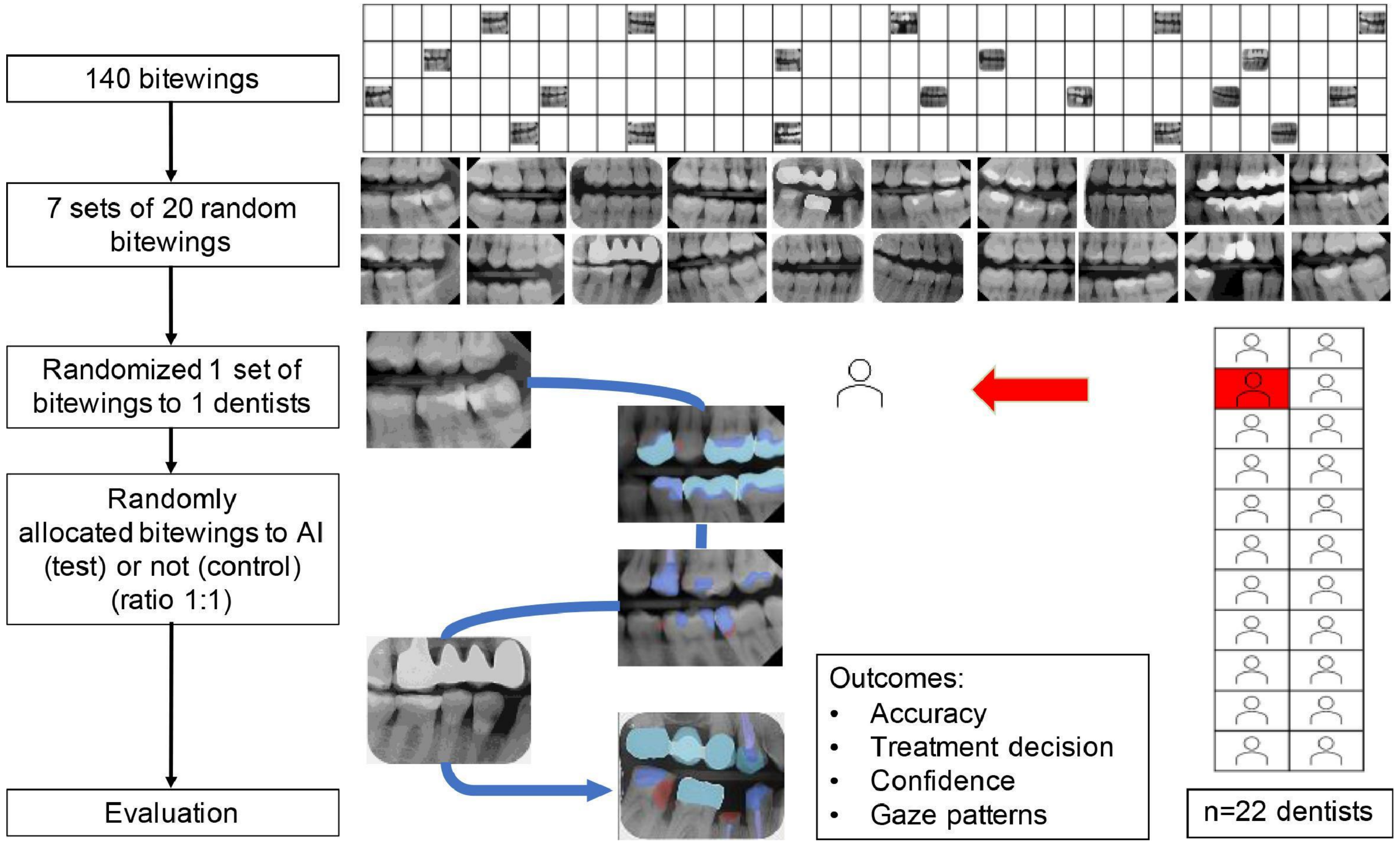
Artificial intelligence for the detection of COVID-19 pneumonia on chest CT using multinational datasets (Harmon et al., 2020)

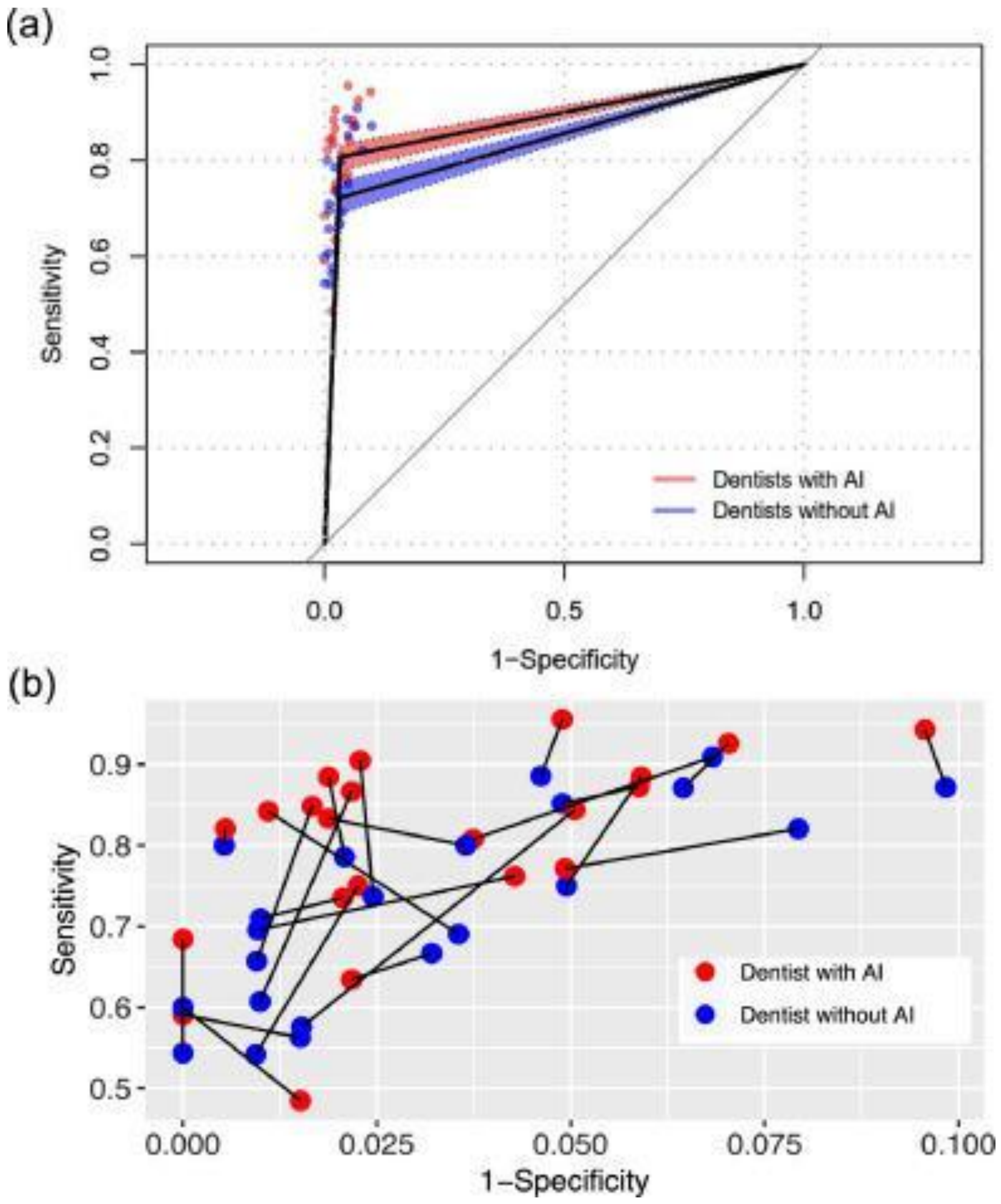


# Diagnosis

## Artificial intelligence for caries detection: Randomized trial (Mertens et al., 2021)



# Benefits and Challenges of AI Integration

# Advantages of AI in Enhancing Patient Care

**Improved Diagnostic Precision**
AI technologies enhance diagnostic precision by utilizing advanced algorithms that analyze imaging data, leading to earlier detection of health issues and more effective treatment strategies tailored to individual patient needs.

**Proactive Patient Management**
Through predictive analytics, AI identifies potential health risks, allowing health professionals to implement preventive measures that reduce the likelihood of severe conditions and associated treatment costs.

**Enhance Patient Accessibility**
AI-driven tele-medicine expands access to care, enabling remote consultations and follow-ups, particularly beneficial for patients in rural or underserved areas, thus improving overall patient engagement and satisfaction.

# Potential Obstacles to AI Adoption in Medicine

**Data Privacy Regulations**
Compliance with strict healthcare data privacy laws, such as HIPAA and GDPR, is critical for medical institutions. Failure to safeguard patient data can lead to legal implications, breaches of confidentiality, and loss of public trust, significantly slowing AI implementation in healthcare.

**Algorithm Bias Risk**
AI systems may exhibit biases if trained on non-representative datasets, leading to inaccurate diagnoses. Continuous monitoring and diverse data inclusion are necessary to ensure equitable and safe AI applications in healthcare.

**High Initial Investment**
The financial burden of acquiring AI technologies, including software and training, can deter smaller practices. A clear demonstration of long-term benefits and ROI is crucial to encourage adoption among these providers.

# Thank You!

## Stay in Touch

**Email: nazila@aimss.ca**